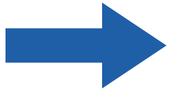
Normalización, transformación y análisis univariante, con datos reales

Juan Miguel Cano Mazón
Unidad de Metabolómica HUMV-IDIVAL
6 de noviembre de 2026

Hasta ahora

la teoría

- Preparación de muestras
  extracción, blancos, QC
- Método de adquisición
  cromatografía, ionización, MS/MS
- Control de calidad
  orden de inyección, estándares internos



Ahora

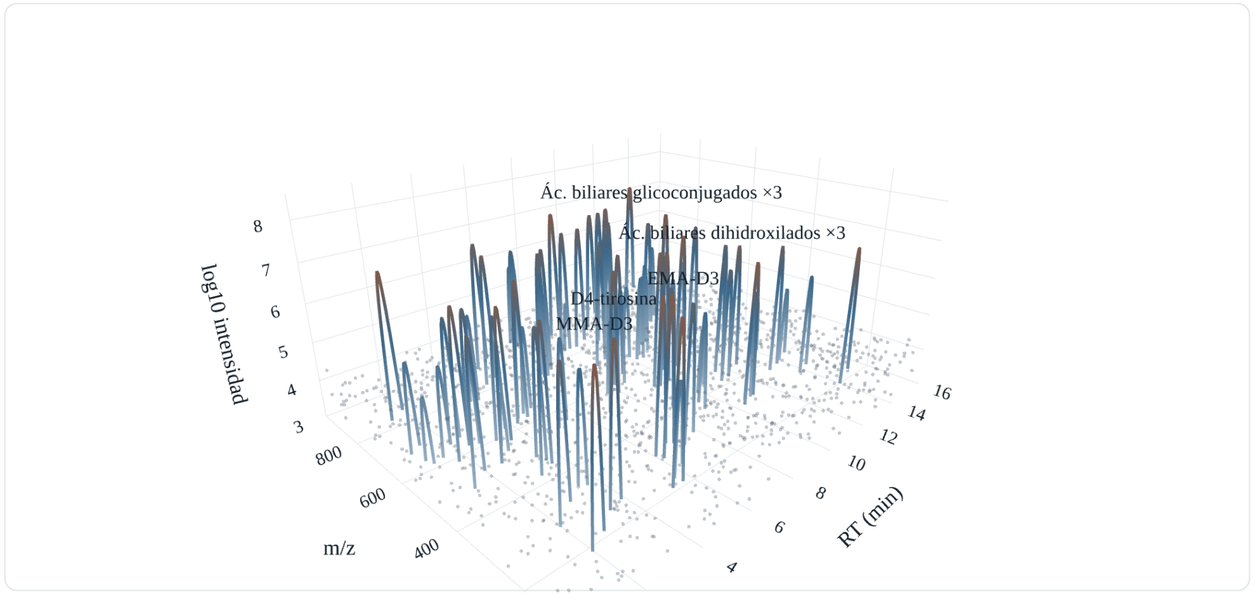
con datos reales, no de libro

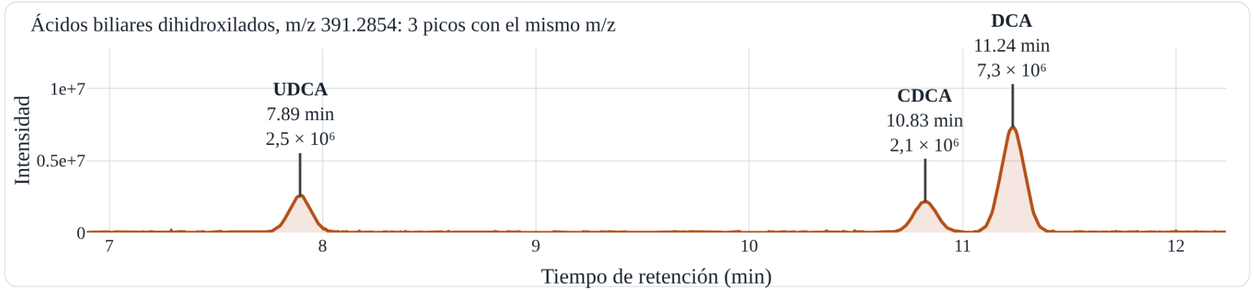
Cómo condicionan los resultados y la forma de proceder

- la preparación de la muestra
- el método de adquisición
- los controles de calidad

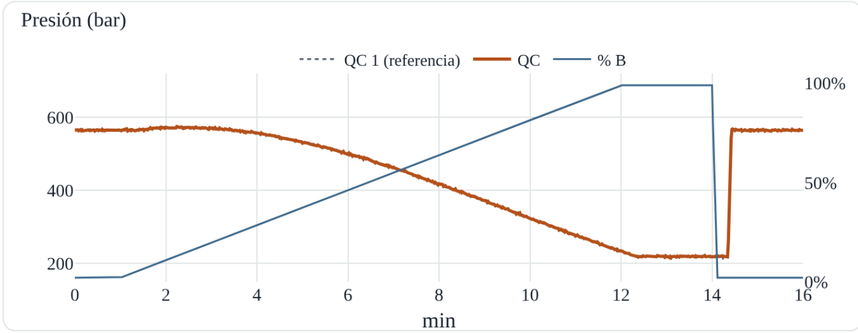
Estructura de los datos LC-MS

Tiempo de retención × m/z × intensidad





Lo que el fichero guarda además (un QC)



Fichero	EPOC_NEG_02_QC.raw
Adquirido	25/06/2026 17:51
Posición del vial	B:A2
Volumen inyectado	3 µL
Temperatura de columna	20,0 °C
Temperatura del automuestreador	16,0 °C
Presión al inicio	563 bar
Presión máxima	573 bar
Voltaje del spray	−2,80 kV
Temperatura del capilar	320 °C
Gas envolvente / auxiliar	40 / 10
Resolución (MS1)	70 000
Objetivo AGC	1e6





La canción

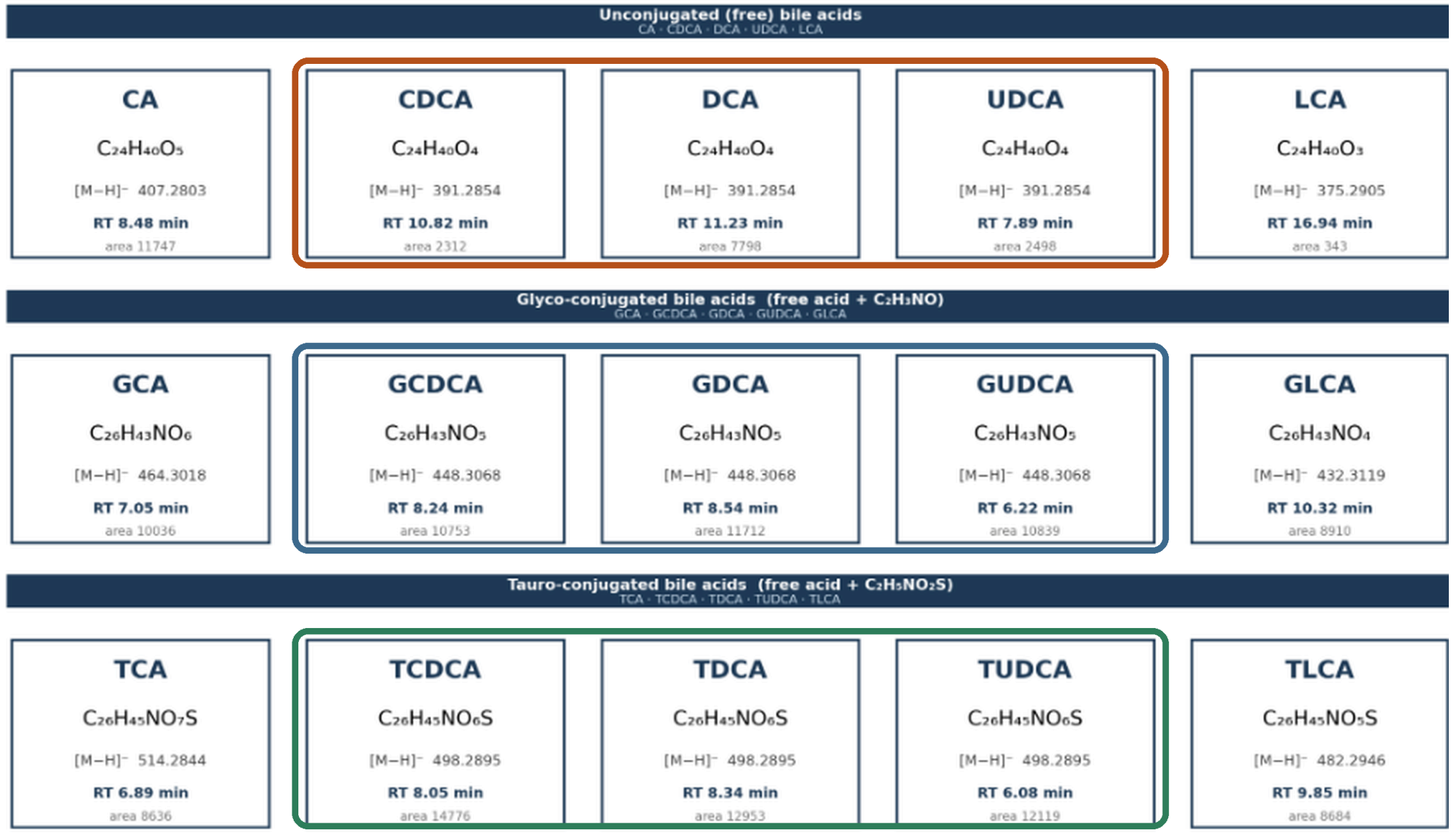
lo que suena = lo que se mide



Sus metadatos

Despacito, Luis Fonsi
Vida, 2019, 3:49, pop latino

Patrones de ácidos biliares



Un m/z puede corresponder a varios compuestos

Por ejemplo, estos isómeros: misma fórmula y masa, distinto tiempo de retención

391,2854: CDCA, DCA y UDCA

448,3068: GCDCA, GDCA y GUDCA

498,2895: TCDCA, TDCA y TUDCA

Conversión de .raw a mzML

.raw
formato del fabricante

→

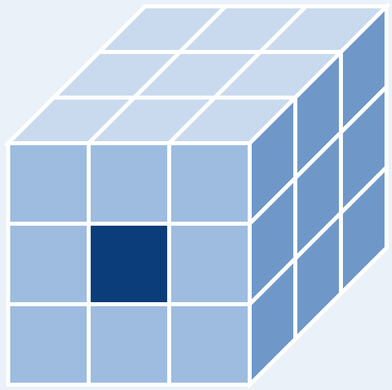
msConvert
ProteoWizard

→

.mzML
estándar abierto, XML

→

.mzMLb
lo que viene



Se lee solo el bloque que hace falta

mzMLb

Mismos metadatos que el mzML, datos en HDF5: bloques comprimidos, como los cubos de meteorología

mzML comprimido

mzMLb, hasta un 77% menos

Bhamber RS et al. bioRxiv, 2020

10	8	9	8	7	8
Controles	**GR4**	**GR3**	**GR2**	**GR1**	**GR5**
	Fumadores sin EPOC	EPOC leve estable	Grave, no agudiza	Grave, agudiza, estable	Exacerbación aguda

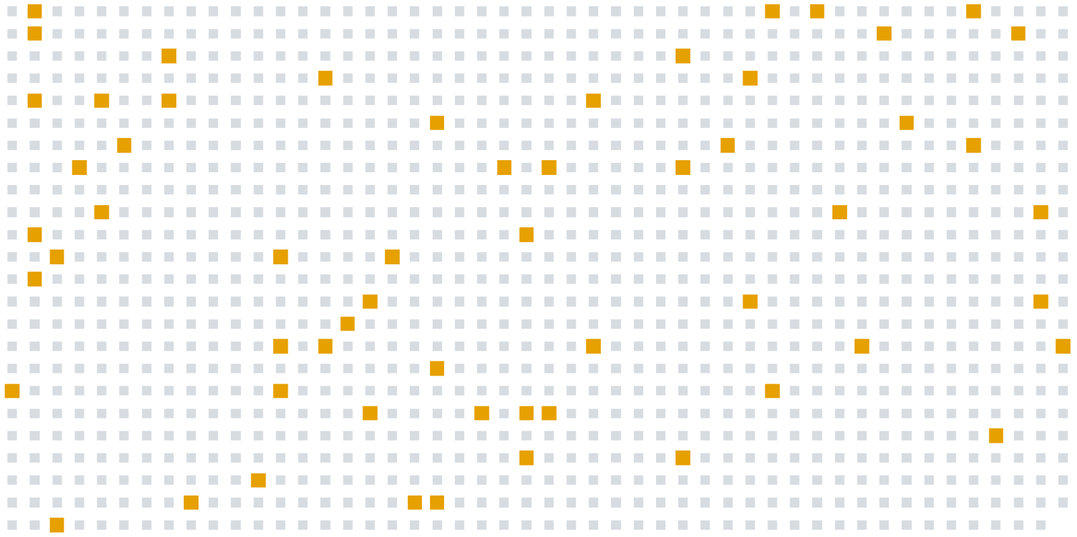
Plasma, LC-HRMS en positivo y negativo, cada muestra inyectada dos veces

Comparaciones múltiples y FDR

Cada cuadrado, un compuesto: 1.151, una prueba para cada uno

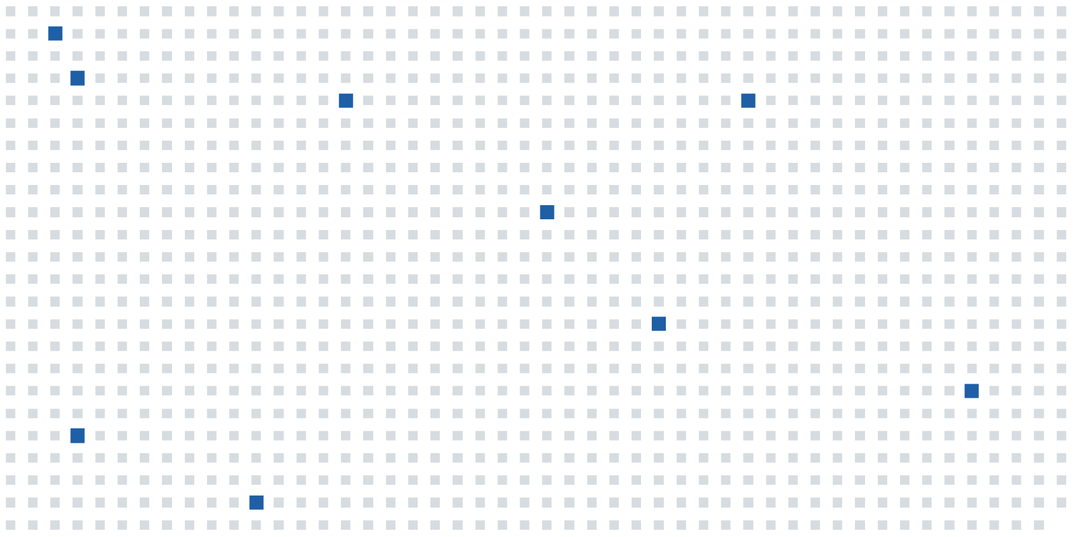
≈ 58 con p < 0,05

los que saldrían solo por azar, sin ninguna diferencia real



9 con FDR < 0,05

los que quedan tras la corrección de Benjamini-Hochberg



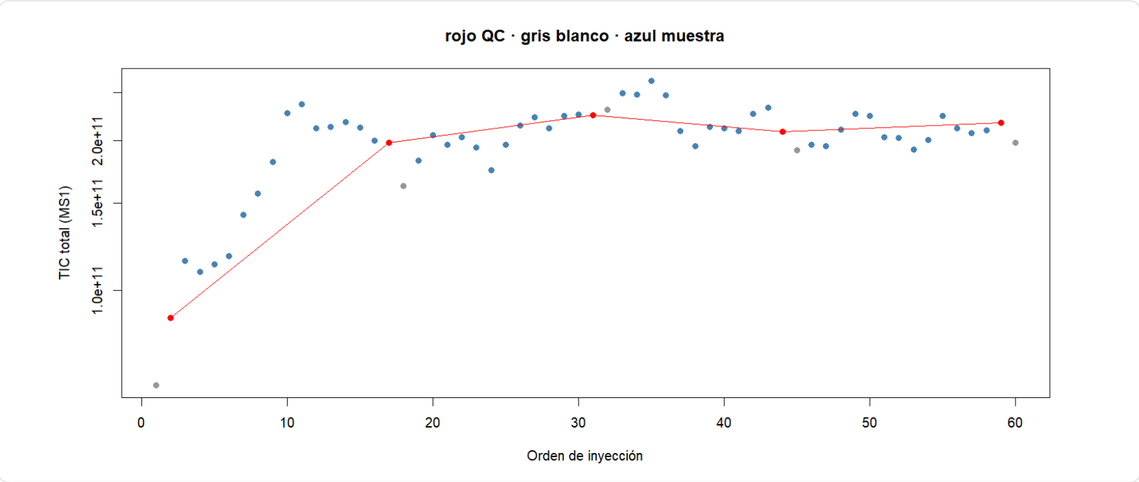
FDR < 0,05: de todo lo que declaro significativo, como mucho un 5% es falso

Réplicas técnicas: significativos por tanda

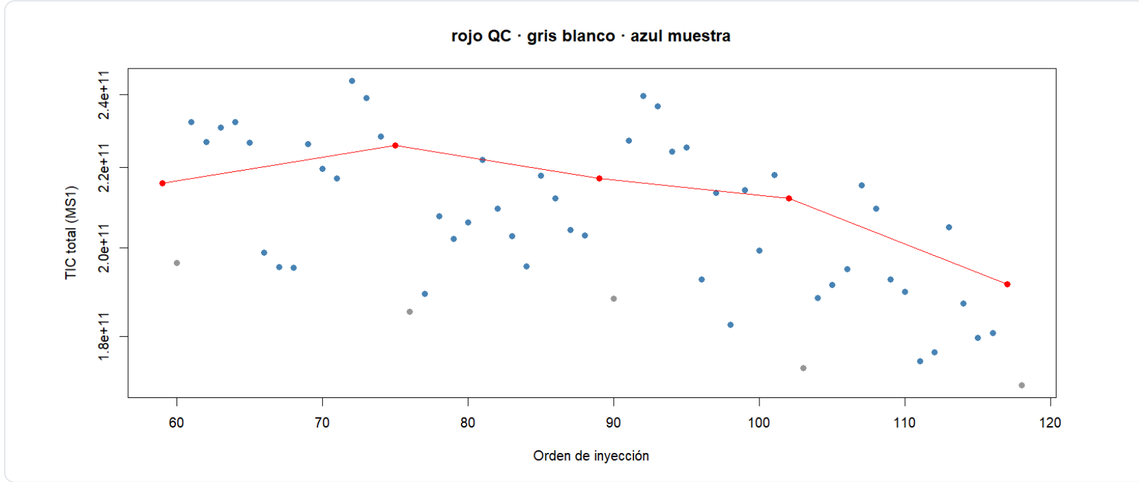
Pareja	Pos 1	Pos 2	Ambas	Se repite	Neg 1	Neg 2	Ambas	Se repite
GR4-GR5	56	75	**34**	61 %	56	73	**32**	57 %
GR2-GR5	36	61	**15**	42 %	38	52	**24**	63 %
GR3-GR5	39	39	**15**	38 %	43	59	**27**	63 %
GR1-GR4	9	13	**2**	22 %	51	33	**16**	31 %
GR1-GR5	14	7	**2**	14 %	0	3	**0**	-
GR1-GR2	2	1	**0**	0 %	9	7	**1**	11 %
GR1-GR3	4	1	**0**	0 %	7	2	**0**	0 %
GR2-GR3	0	4	**0**	-	0	0	**0**	-
GR2-GR4	0	5	**0**	-	2	0	**0**	0 %
GR3-GR4	0	0	**0**	-	0	0	**0**	-

FDR < 0,05

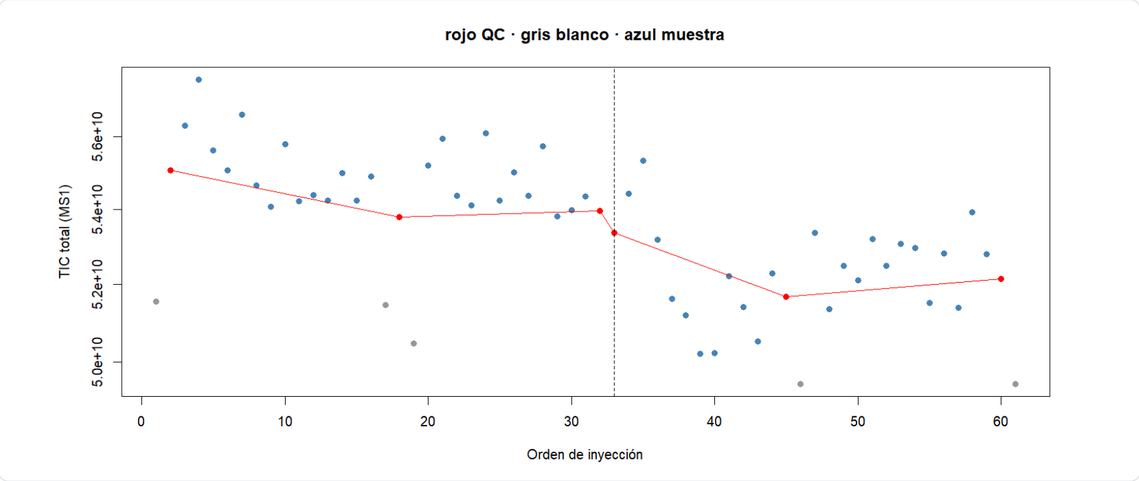
TIC frente al orden de inyección



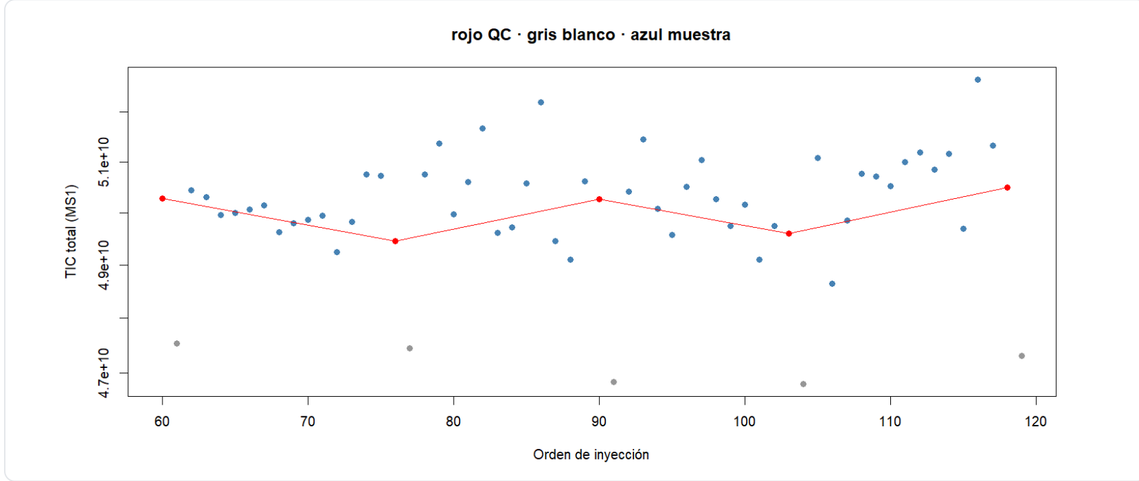
Pos 1



Pos 2

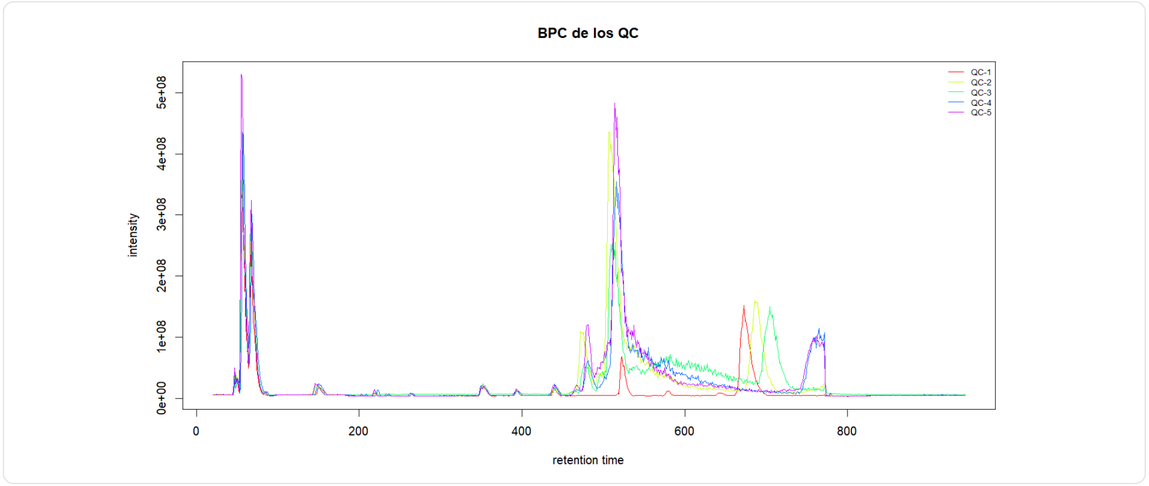


Neg 1

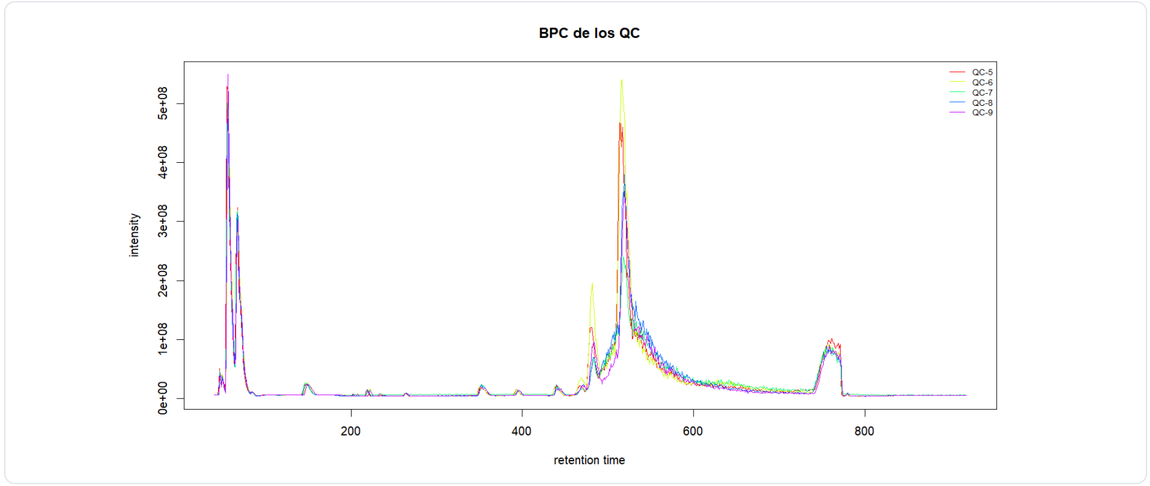


Neg 2

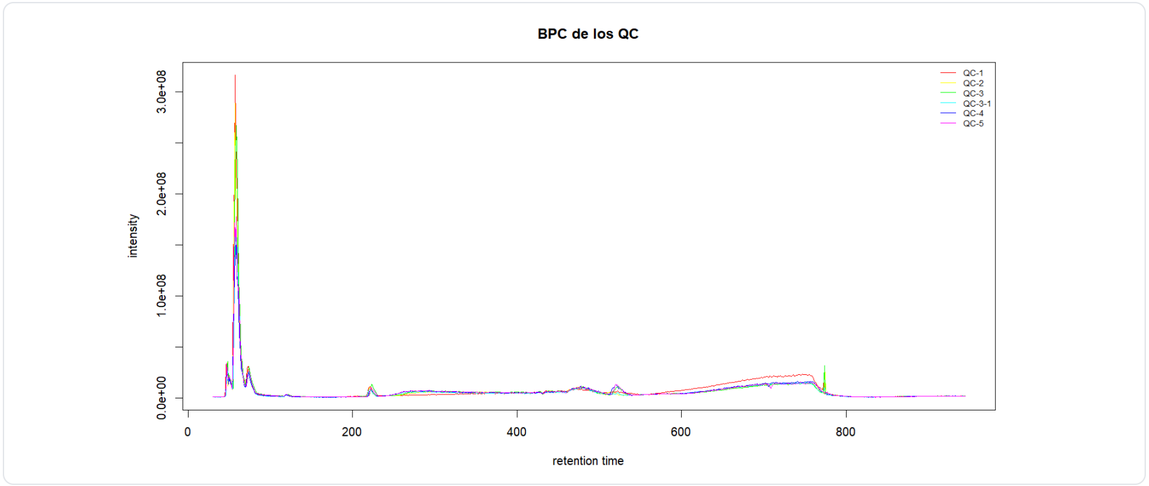
BPC de los QC



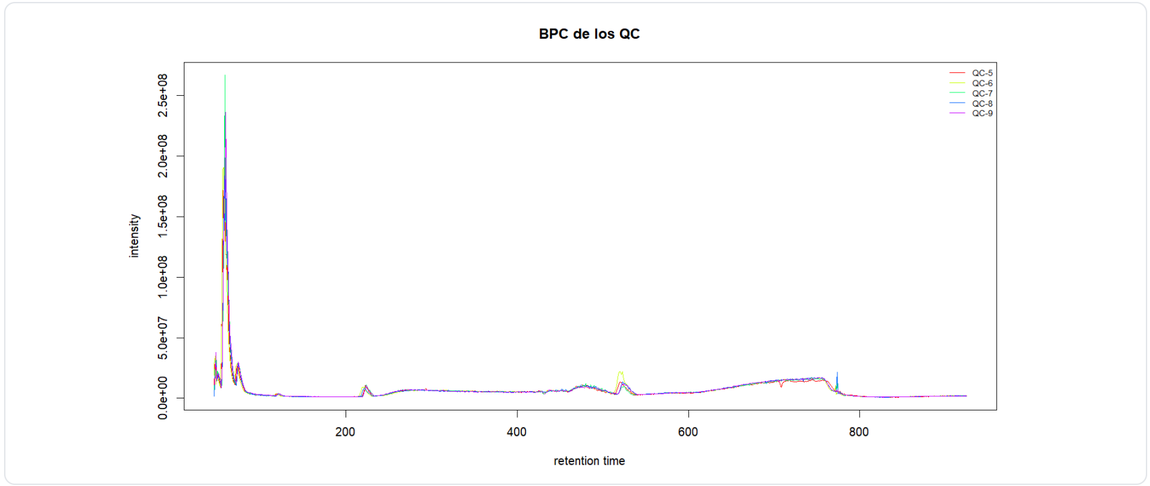
Pos 1



Pos 2

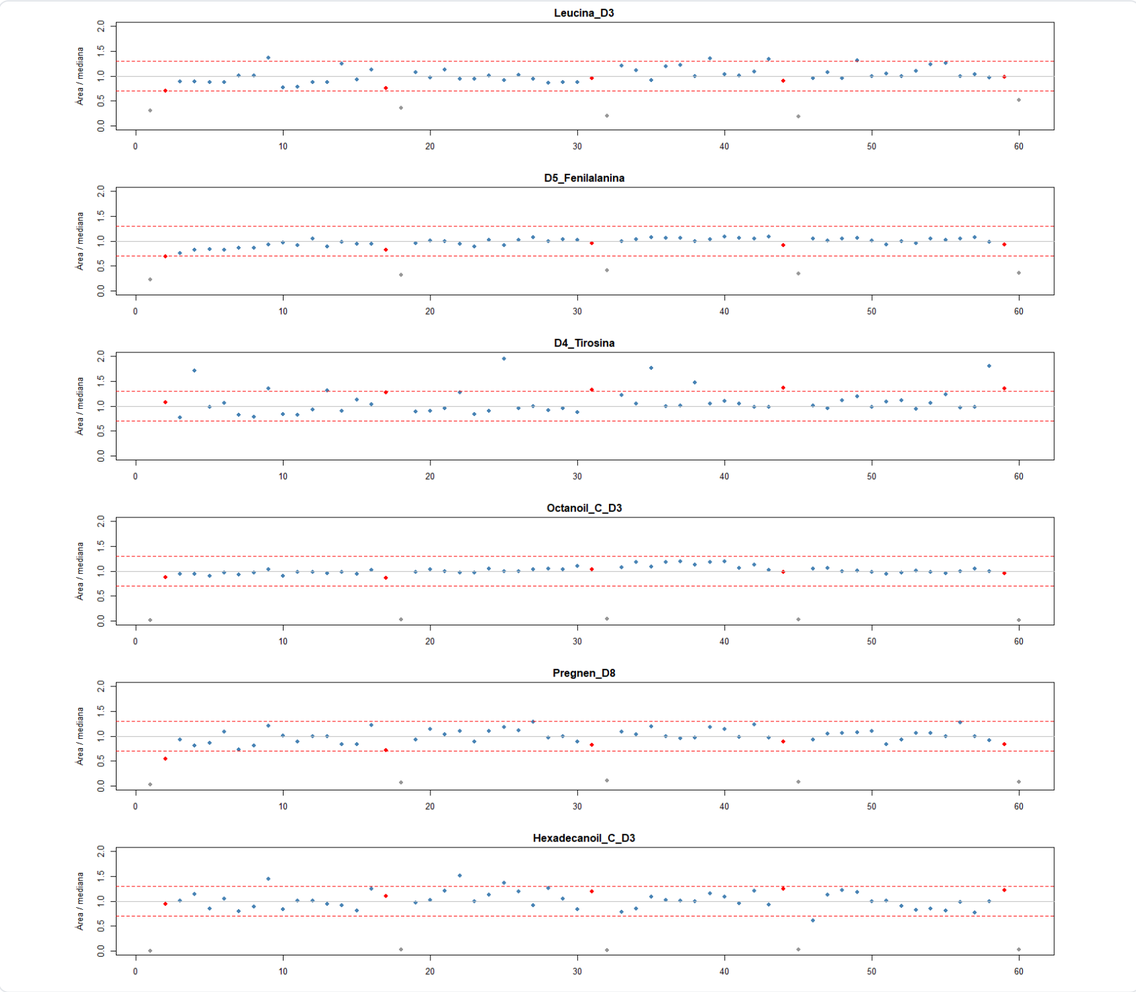


Neg 1

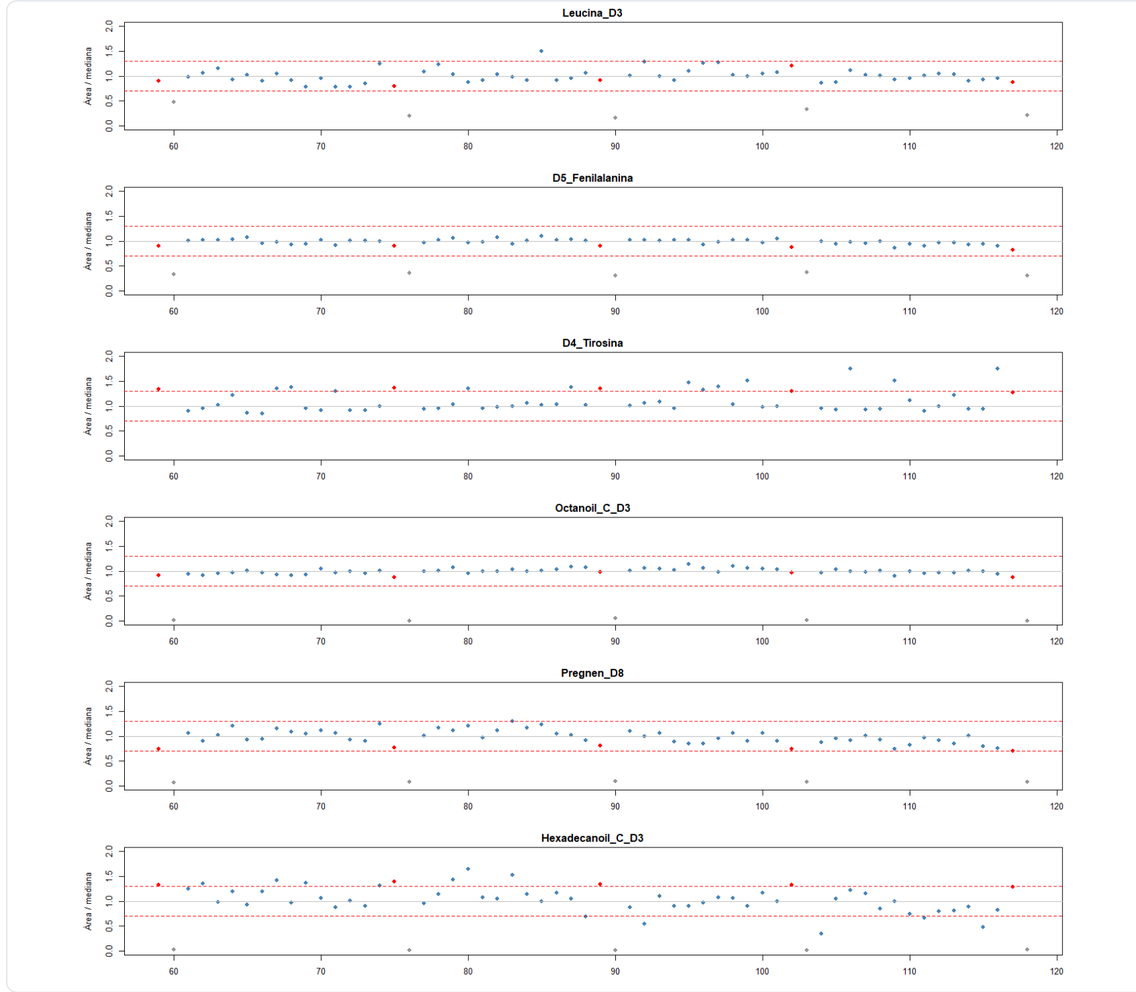


Neg 2

Estándares internos en positivo



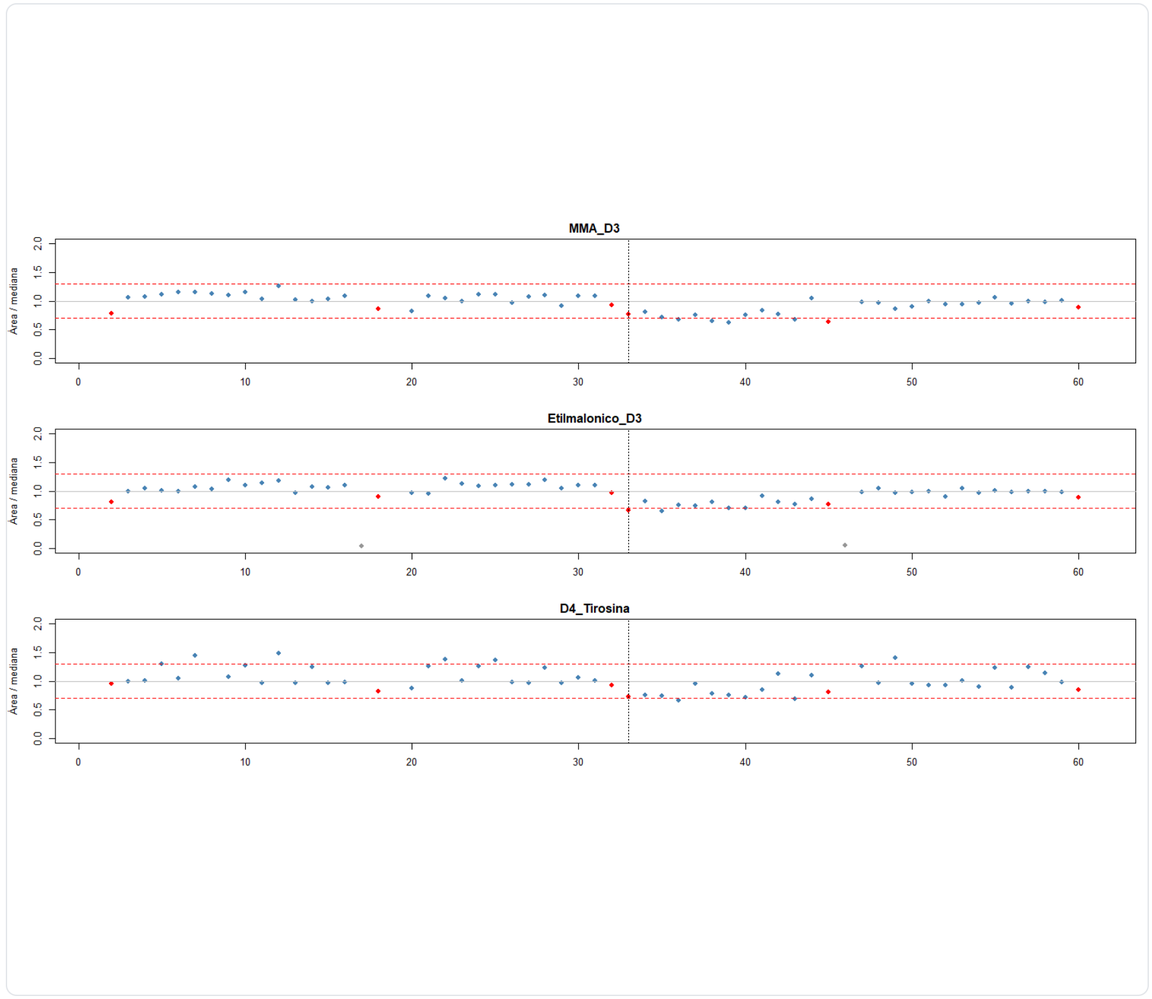
Pos 1



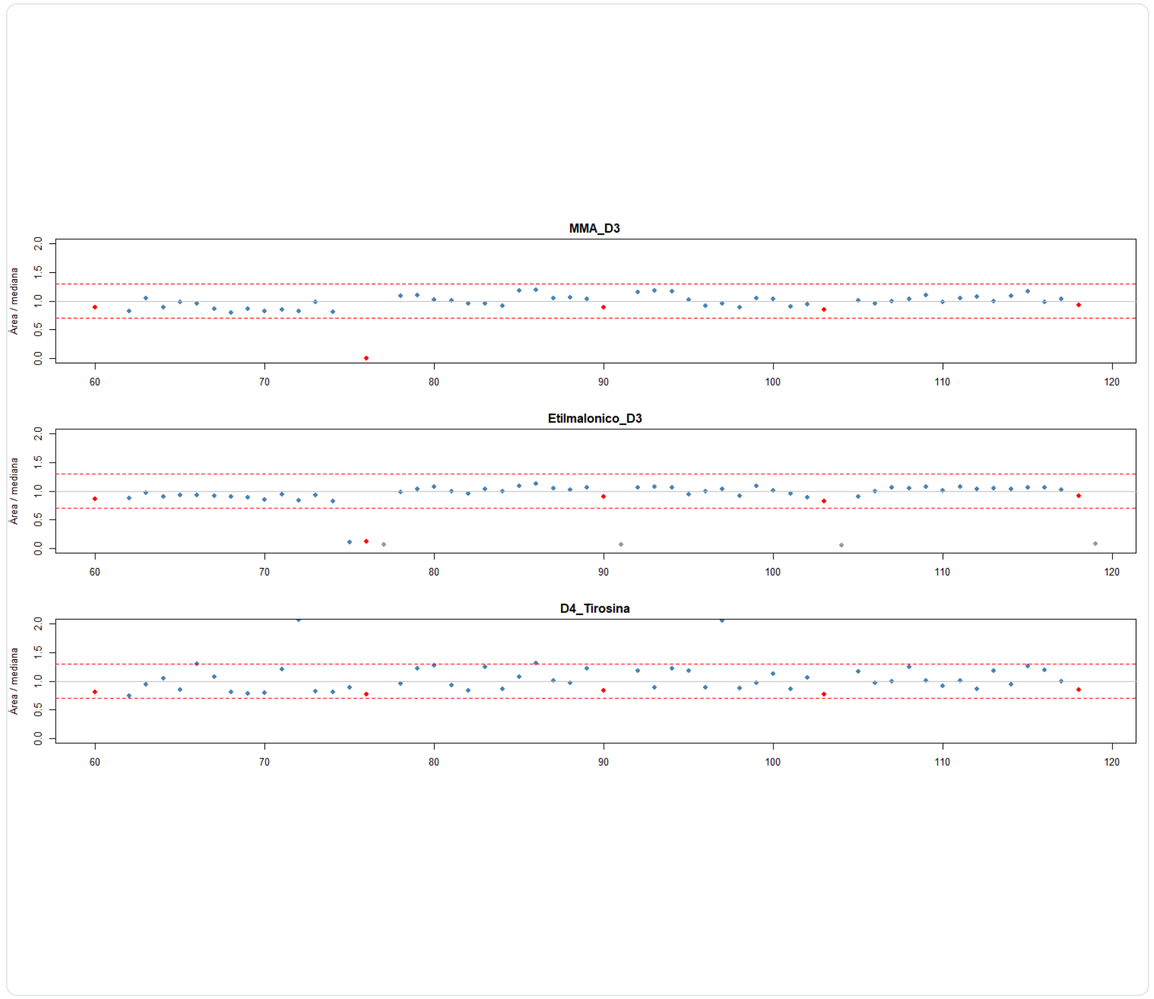
Pos 2

Leucina-D3, D5-fenilalanina, D4-tirosina, octanoíl-carnitina D3, pregnenolona D8, hexadecanoíl-carnitina D3

Estándares internos en negativo



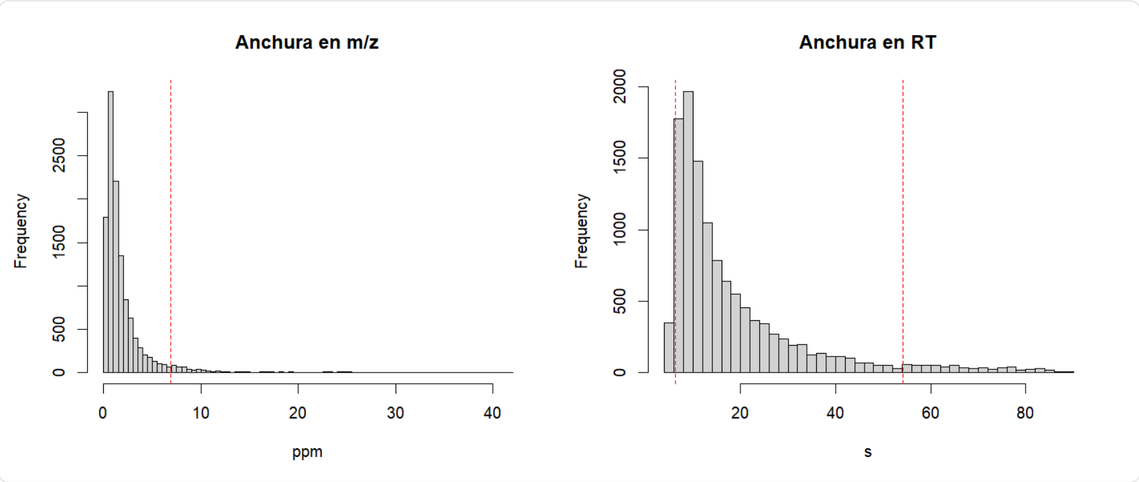
Neg 1



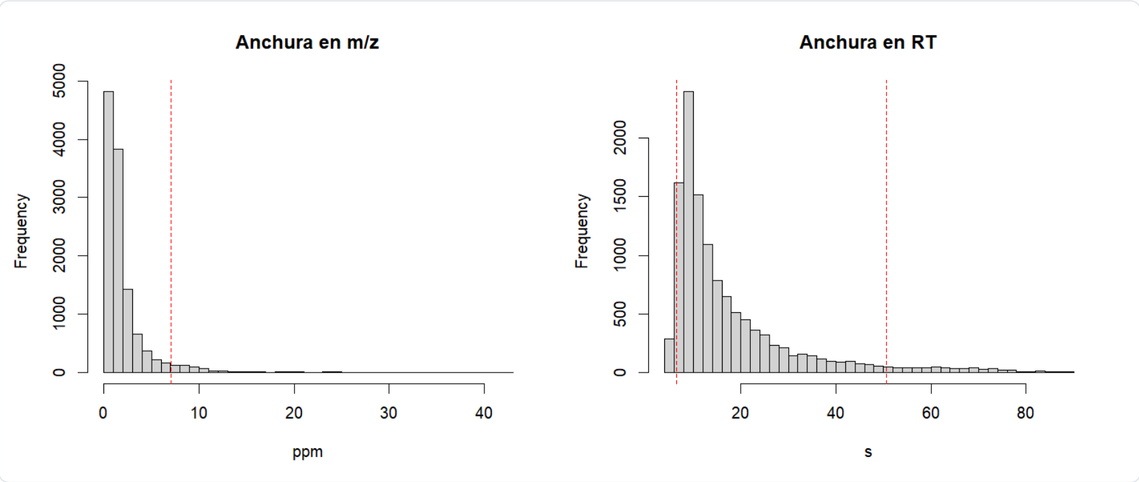
Neg 2

MMA-D3 (76 s), ácido etilmalónico D3 (120 s), D4-tirosina (93 s)

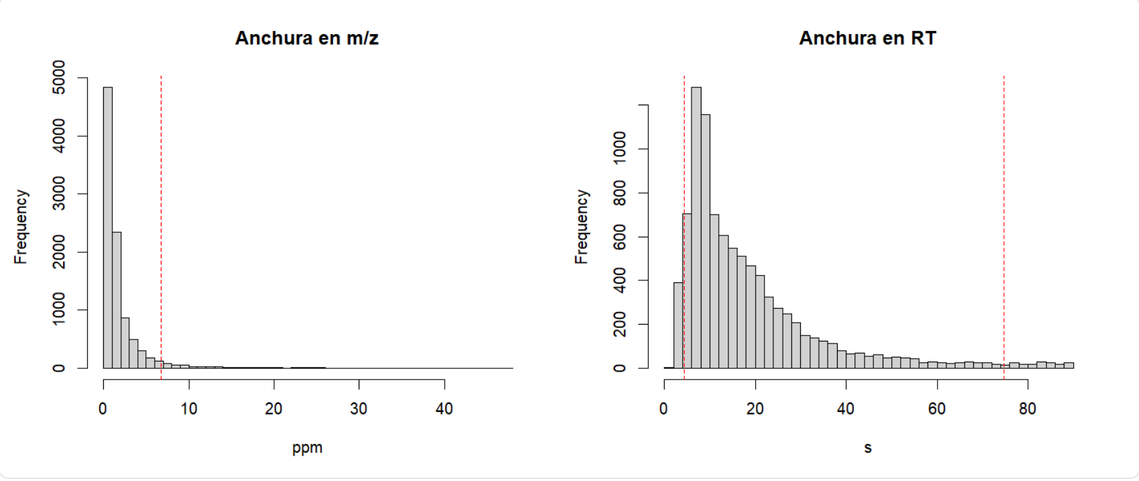
Anchura de los picos en masa y en tiempo



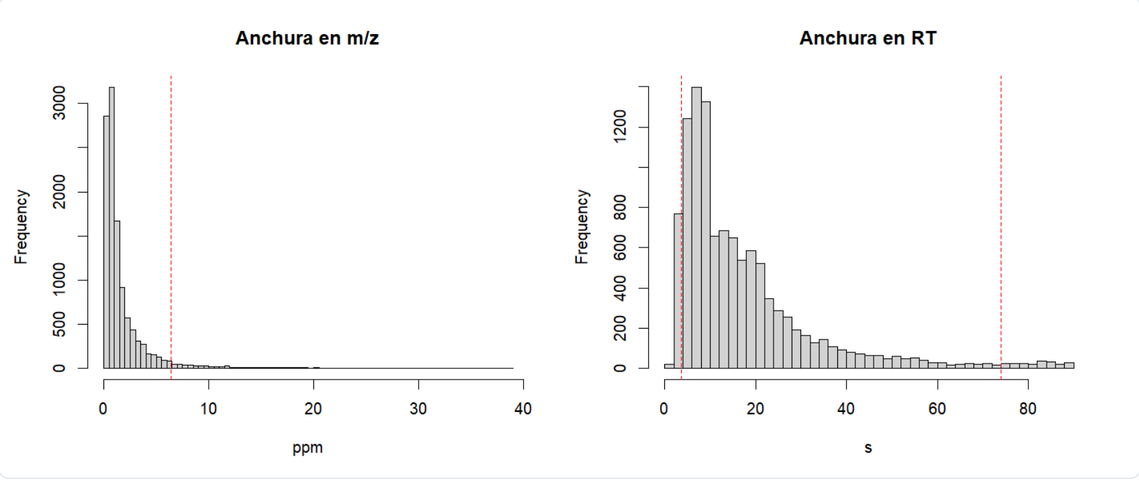
Pos 1



Pos 2

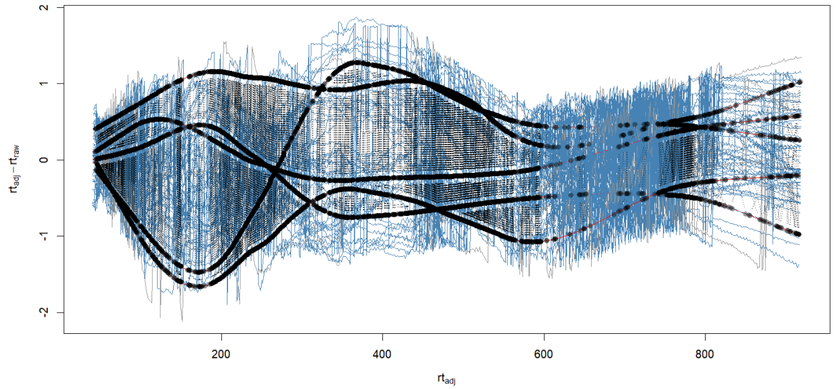


Neg 1

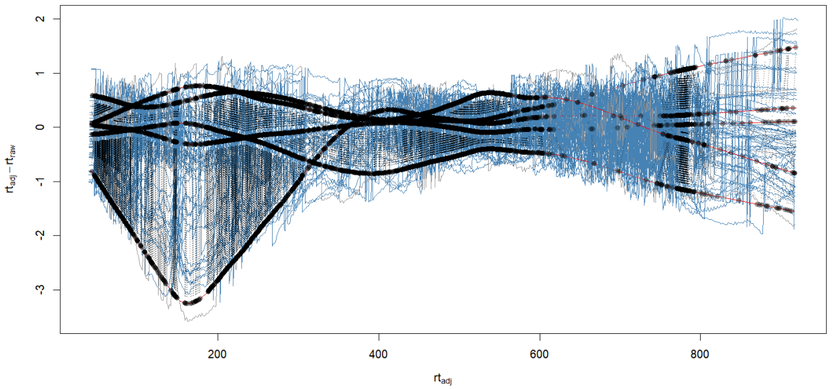


Neg 2

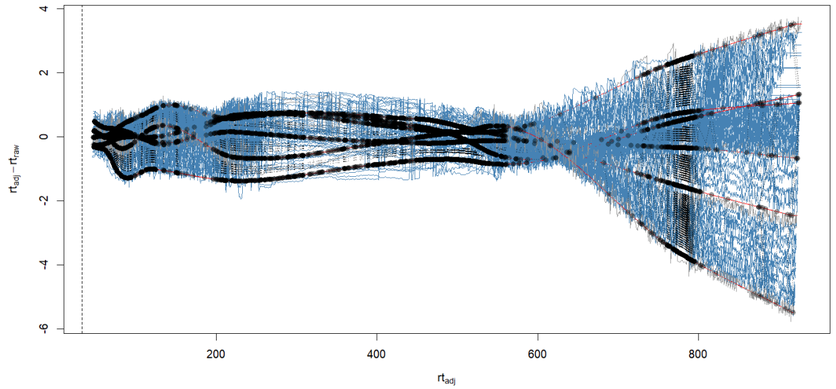
Alineamiento de los tiempos de retención



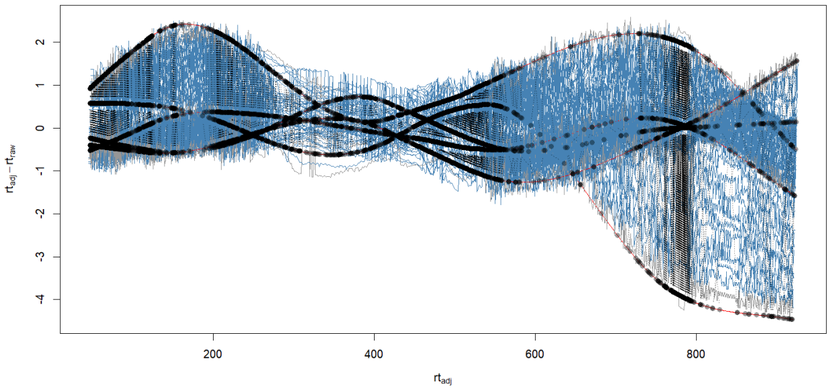
Pos 1: ± 2 s a lo largo de todo el gradiente



Pos 2: hasta 3 s, hacia el minuto 3

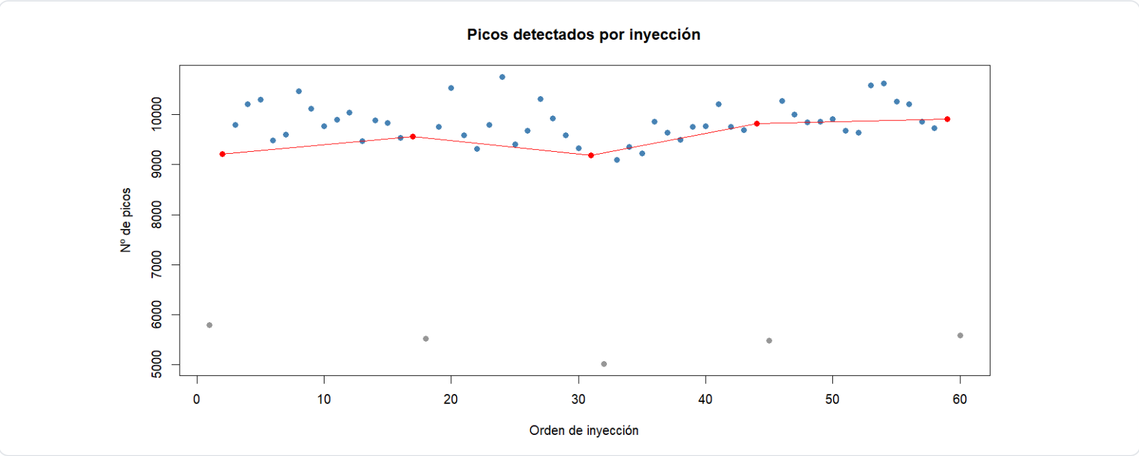


Neg 1: ± 1 s hasta el minuto 10; se abre al final

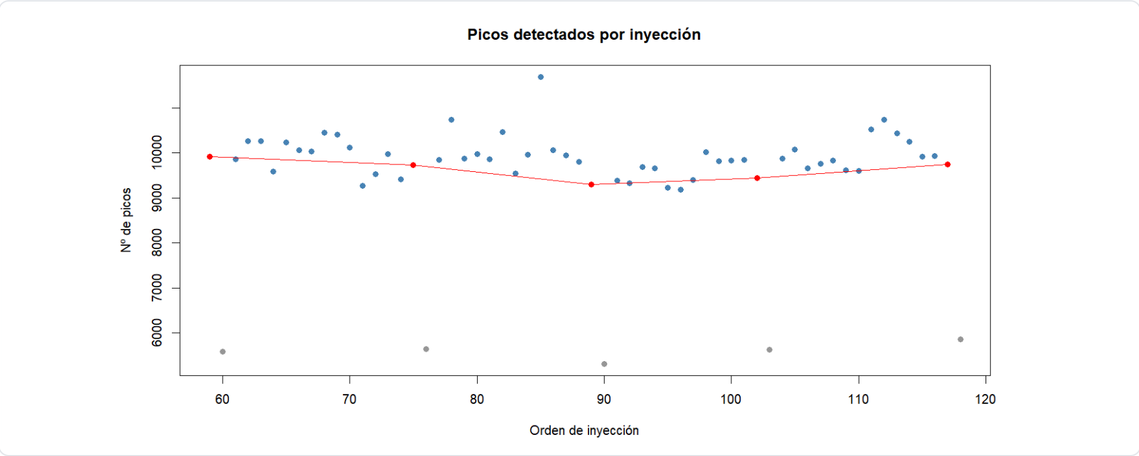


Neg 2: ± 2 s; hasta 5 s al final

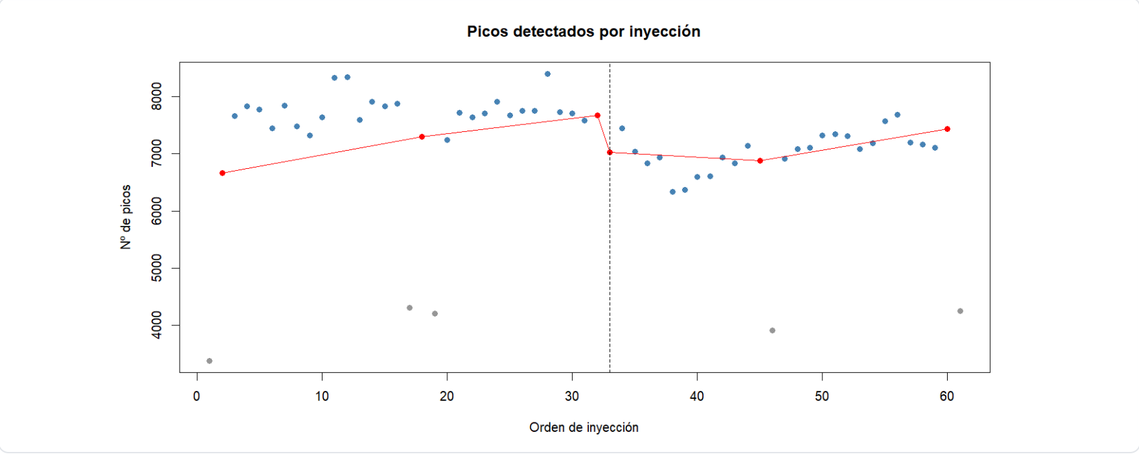
Picos detectados por inyección



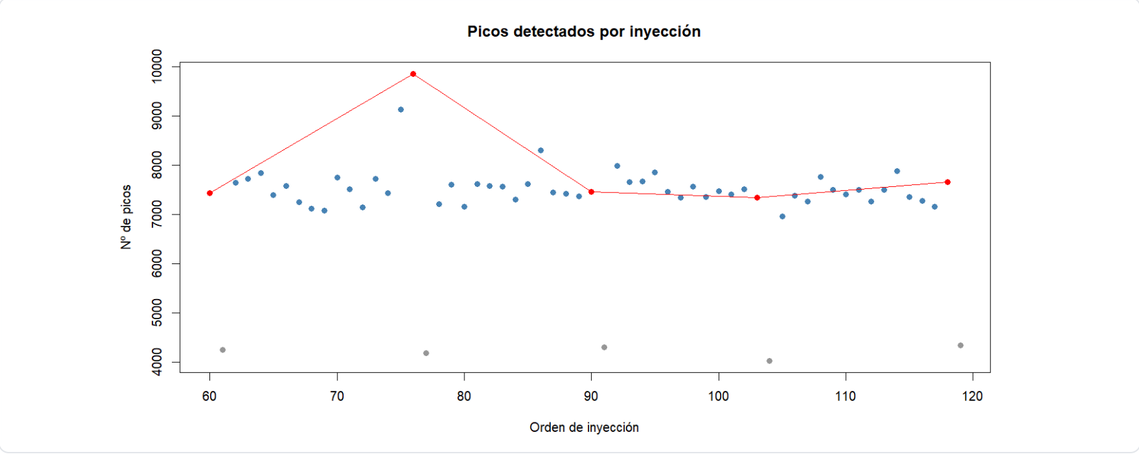
Pos 1: **9.795** muestras, **9.564** QC, **5.523** blancos



Pos 2: **9.872** muestras, **9.728** QC, **5.635** blancos

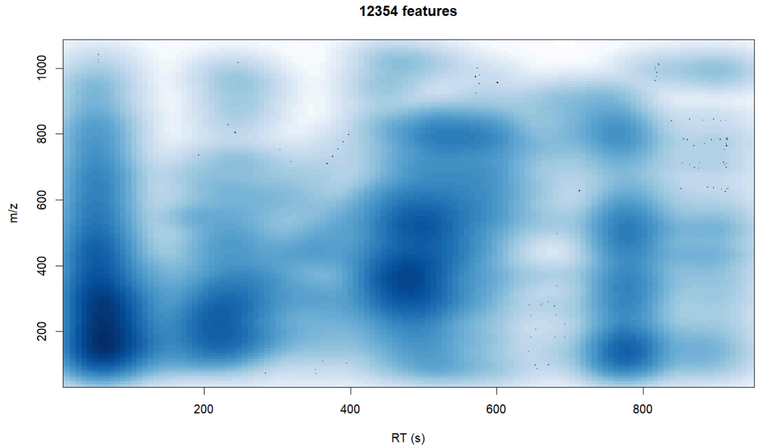


Neg 1: **7.463** muestras, **7.163** QC, **4.205** blancos

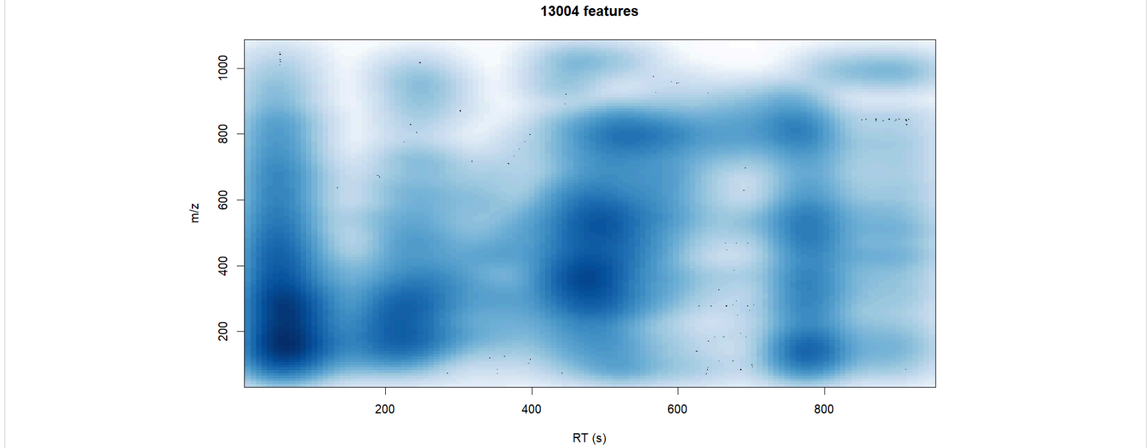


Neg 2: **7.492** muestras, **7.458** QC, **4.251** blancos

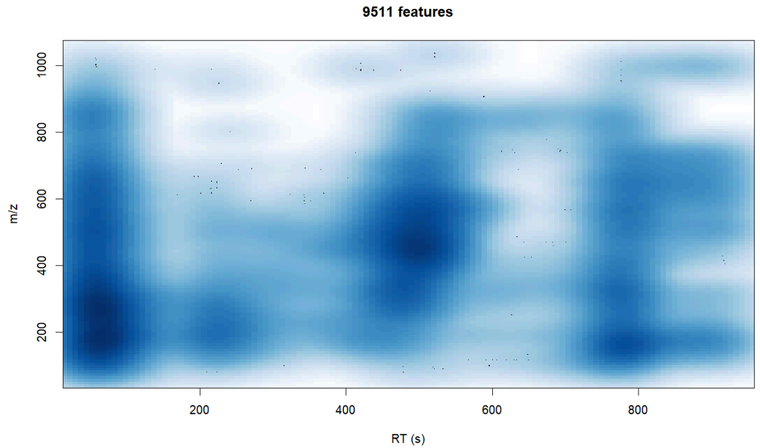
Features en masa frente a tiempo



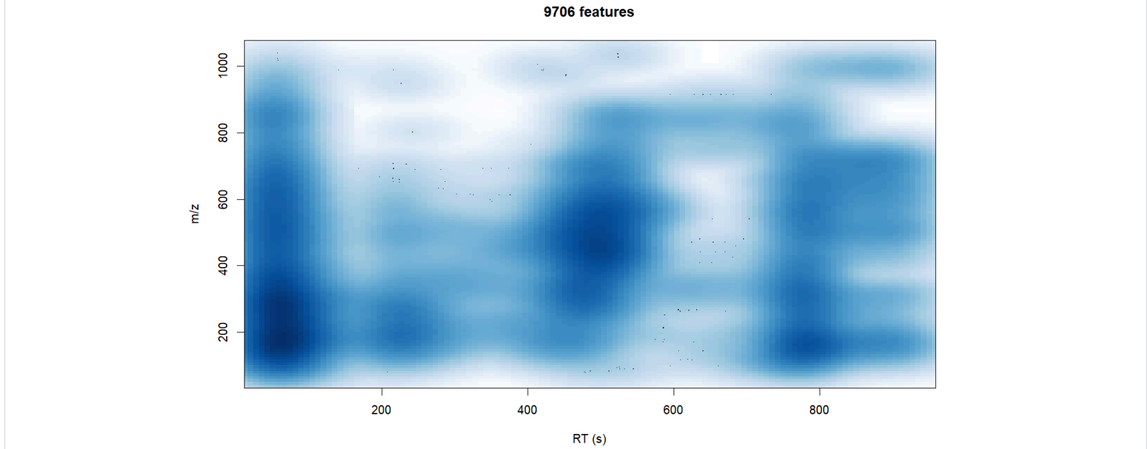
Pos 1, 12.354 features



Pos 2, 13.004

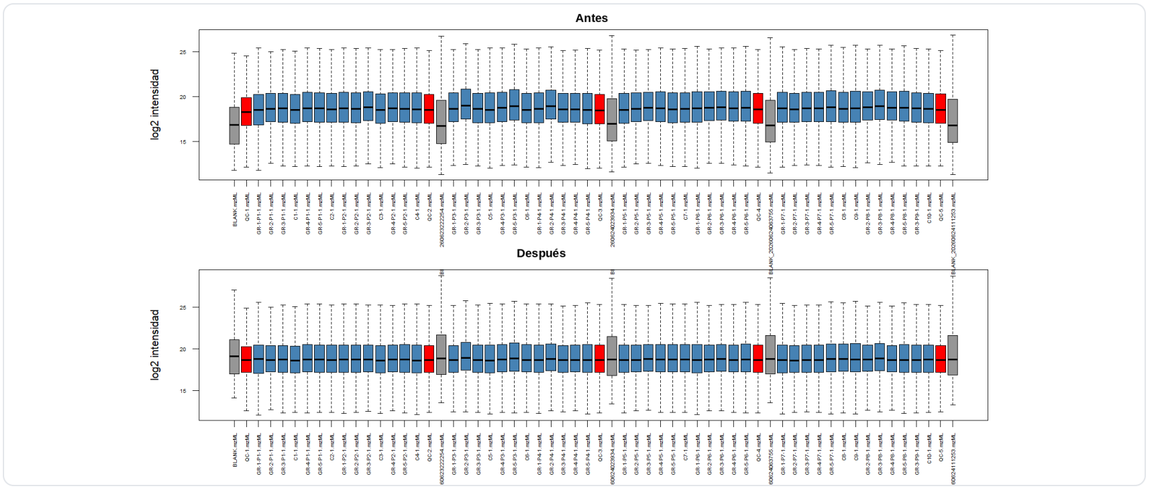


Neg 1, 9.511

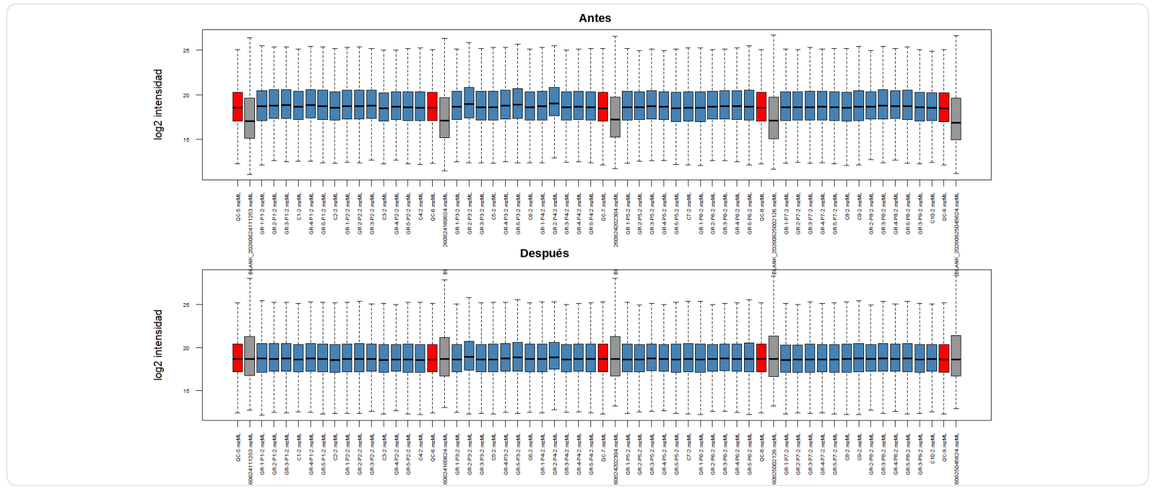


Neg 2, 9.706

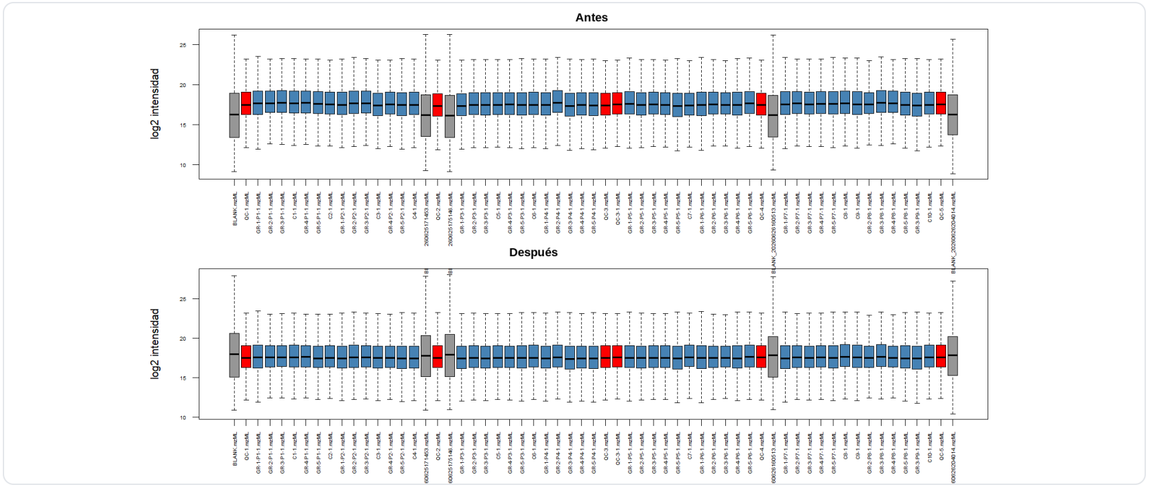
Normalización



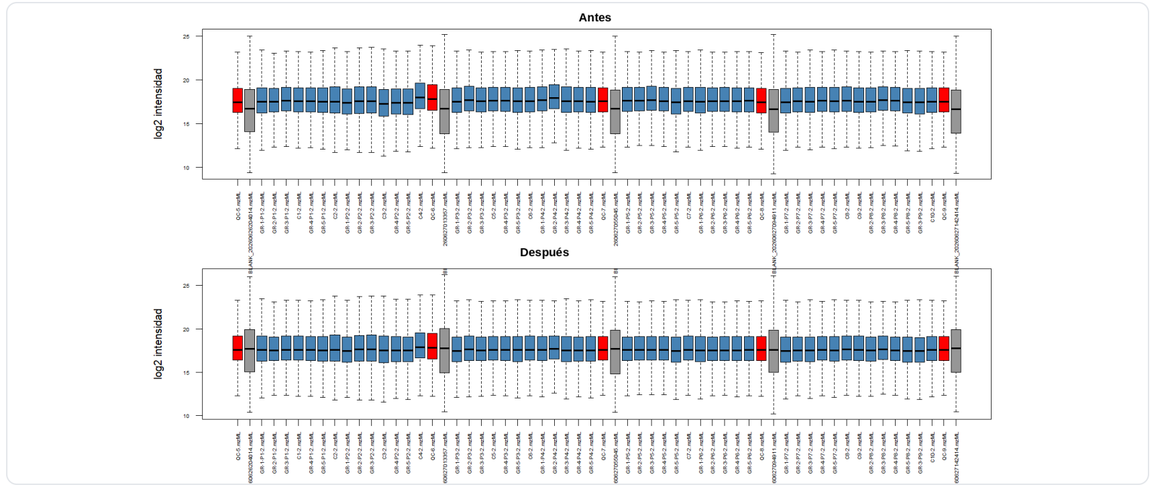
Pos 1



Pos 2

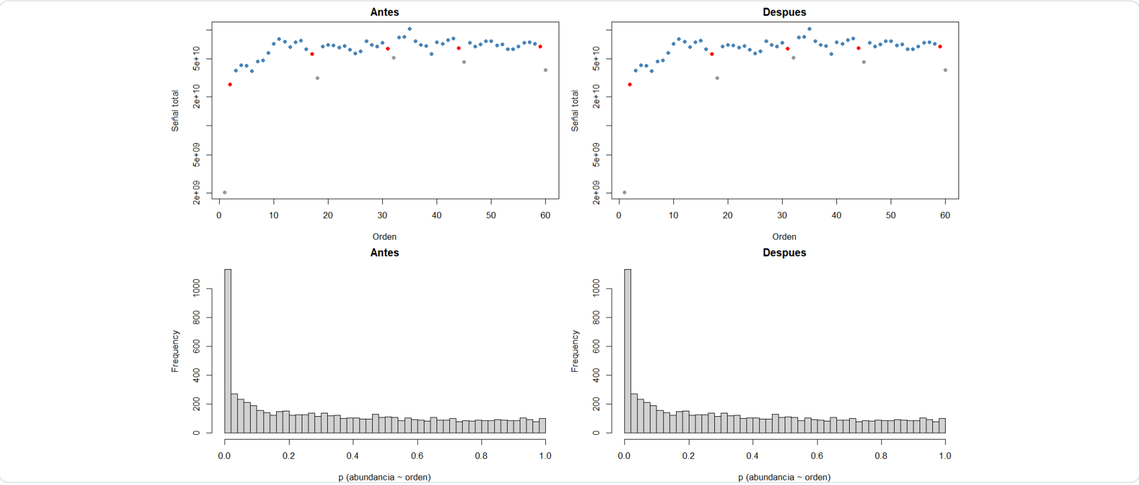


Neg 1

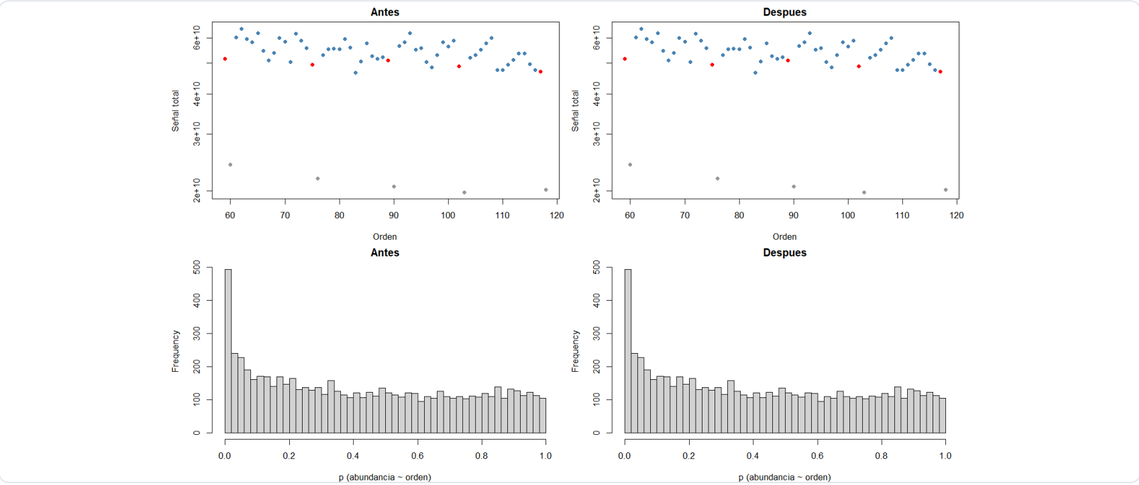


Neg 2

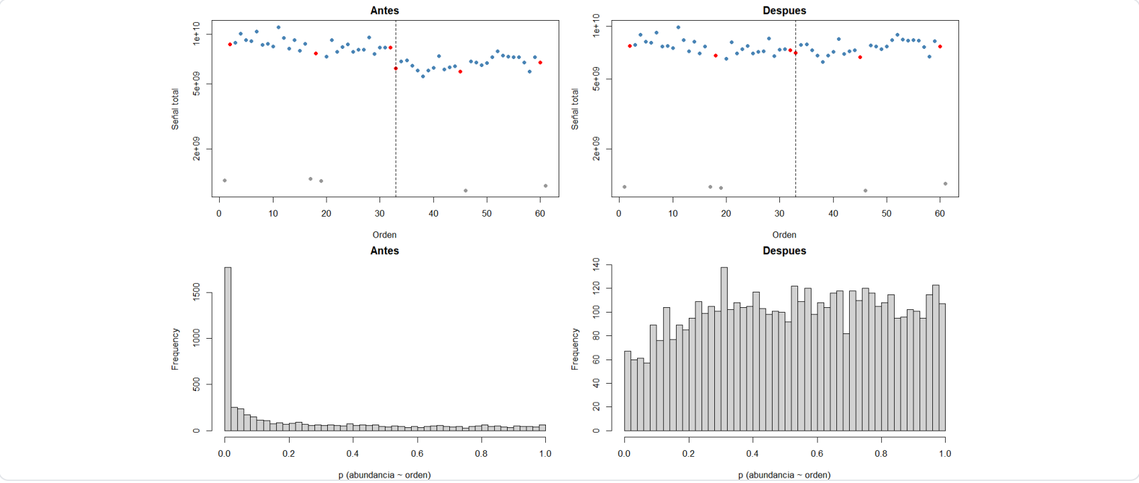
Corrección de la deriva con los QC



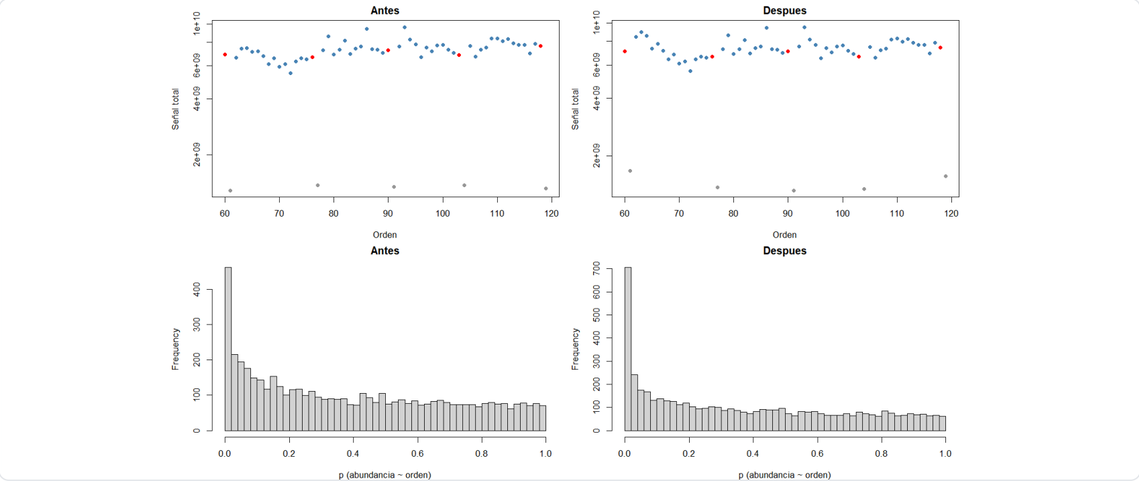
Pos 1



Pos 2



Neg 1, el histograma queda plano

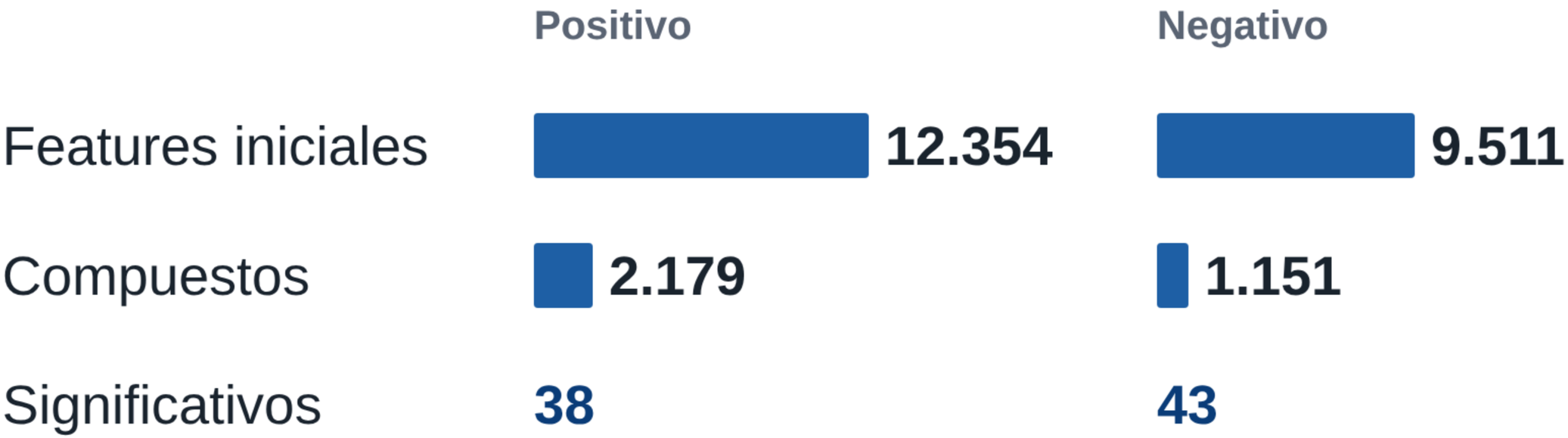


Neg 2

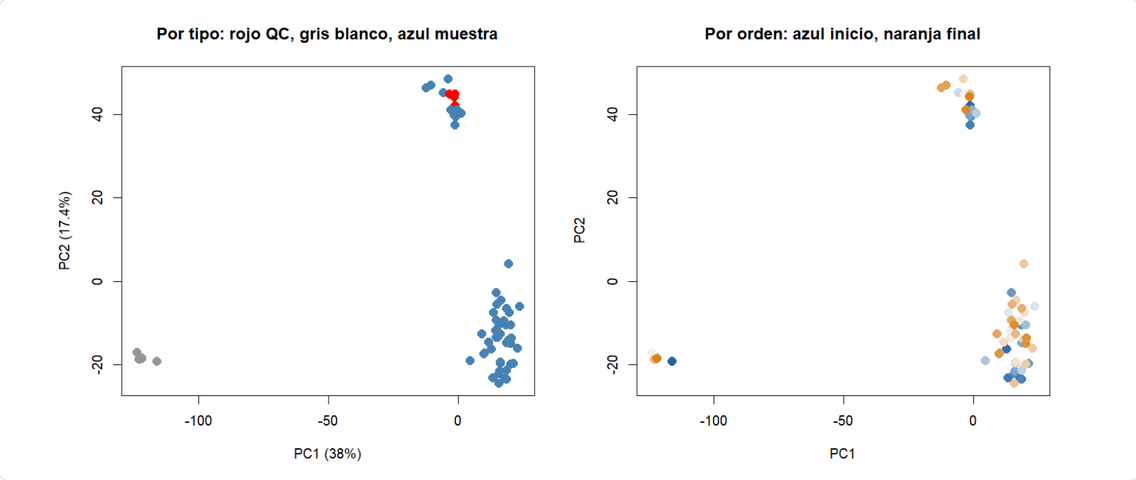
Filtros de existencia y de calidad

	Pos 1	Pos 2	Neg 1	Neg 2
Iniciales	12.354	13.004	9.511	9.706
Baja detección en el grupo	-868	-1.100	-913	-851
Fondo del sistema (blancos)	-4.781	-5.052	-3.553	-3.690
Calidad en los QC	-2.893	-2.487	-2.758	-1.879
Tras calidad	**3.812**	**4.365**	**2.287**	**3.286**
Compuestos, sin aductos ni isótopos	2.179	2.544	1.151	1.747

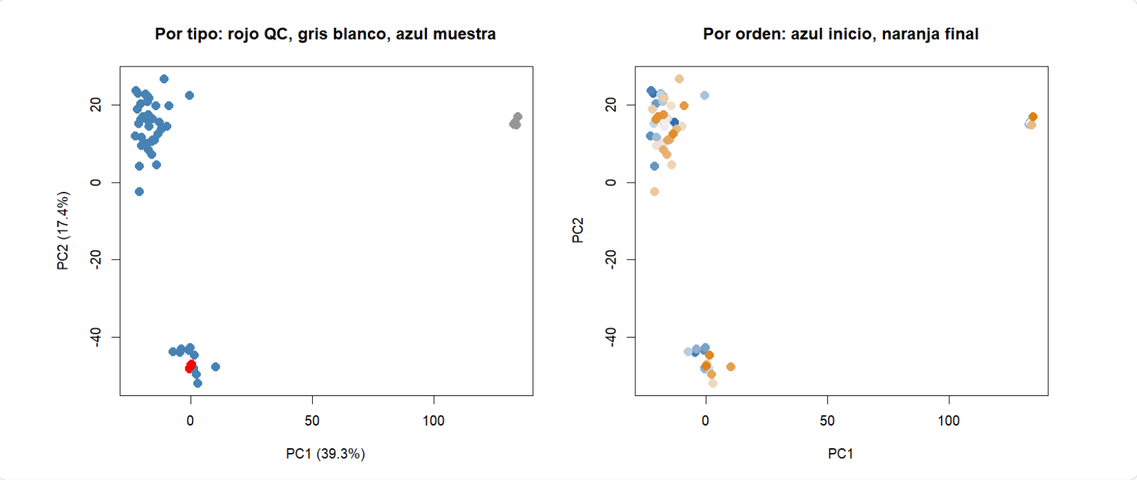
Resumen del procesado



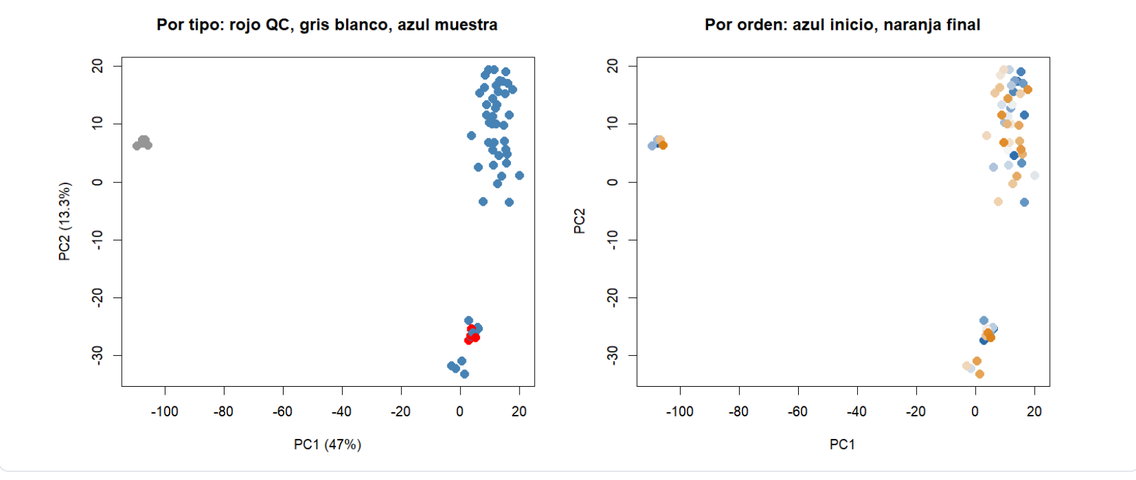
PCA de todas las inyecciones



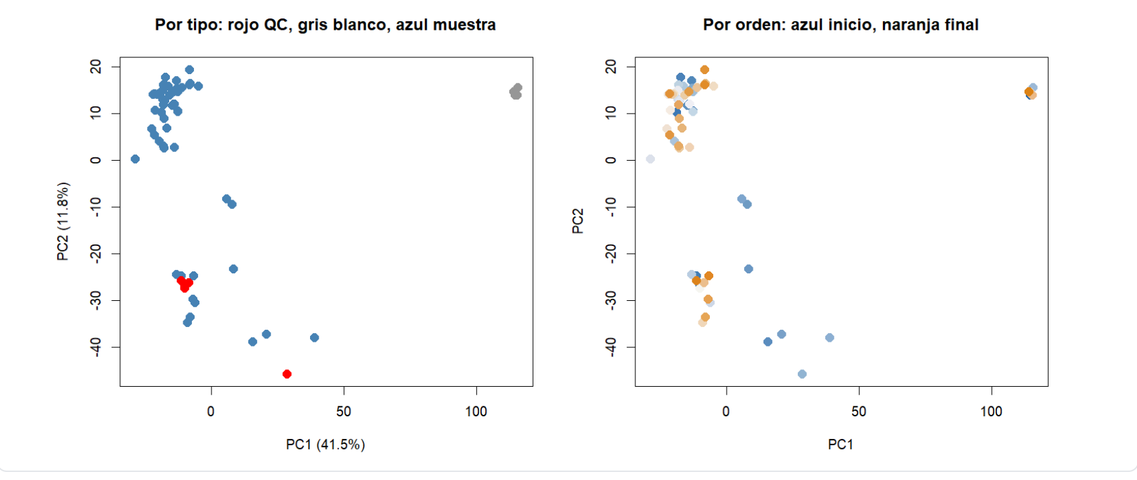
Pos 1



Pos 2

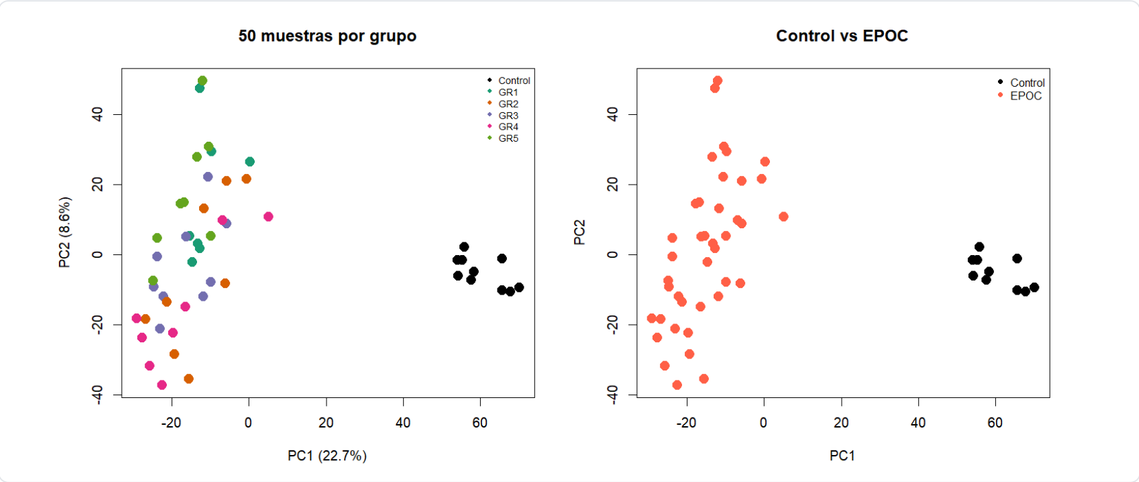


Neg 1

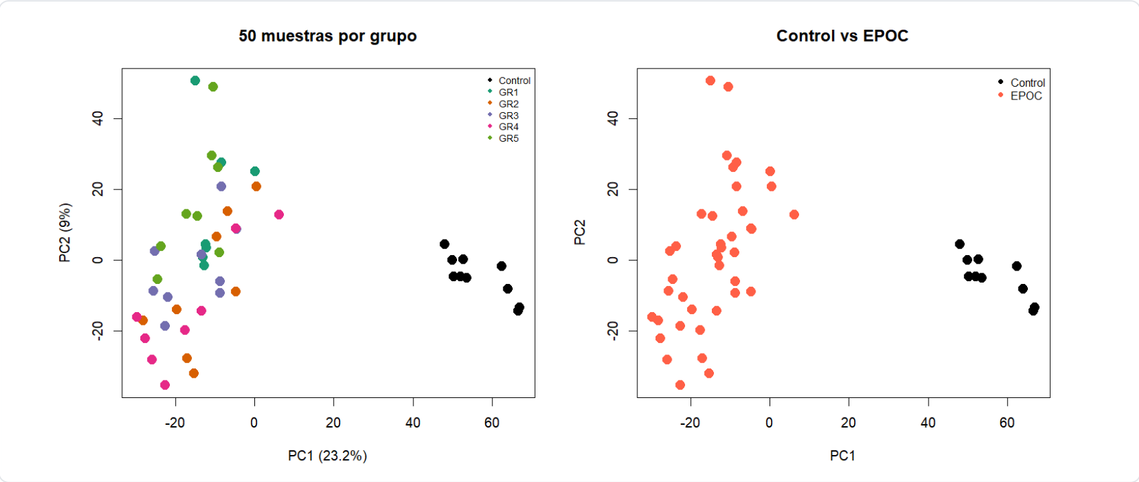


Neg 2

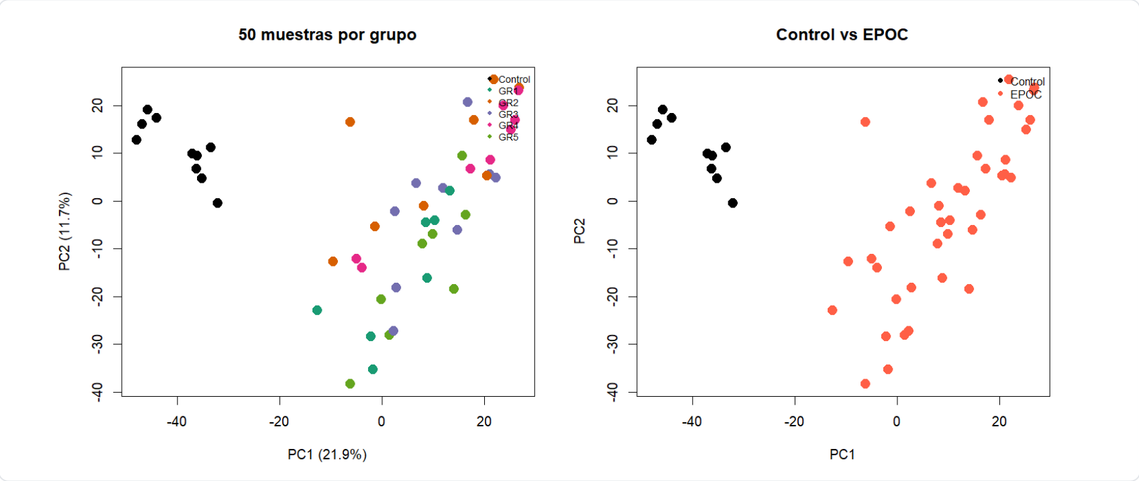
PCA de las 50 muestras



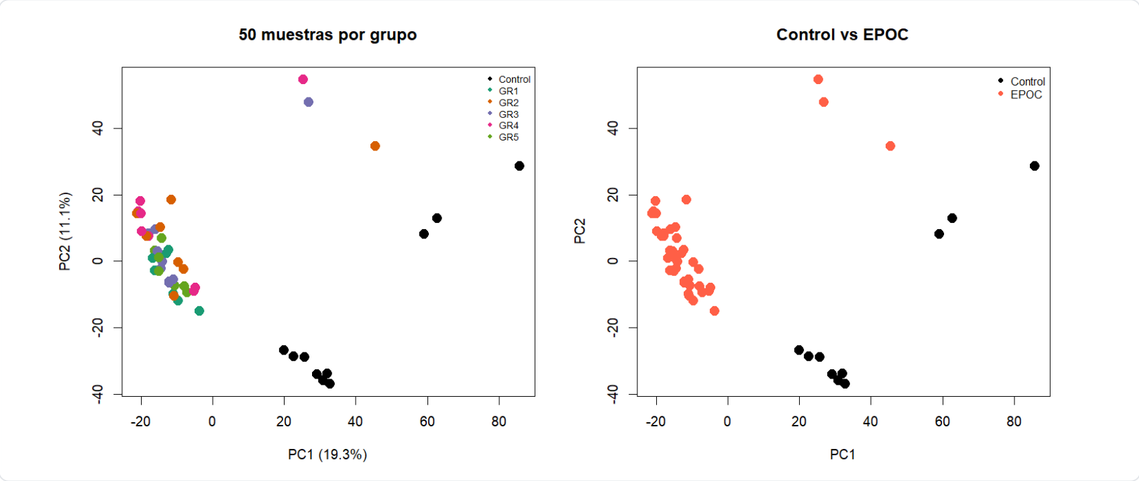
Pos 1



Pos 2

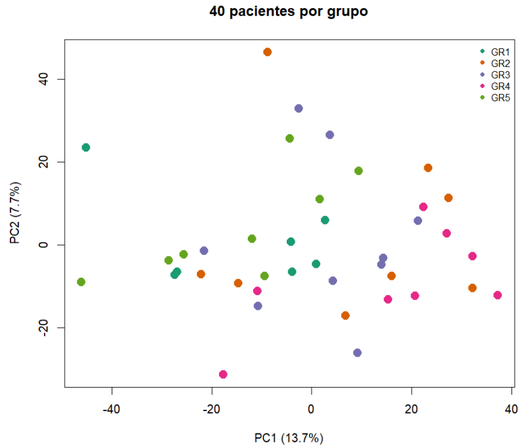


Neg 1

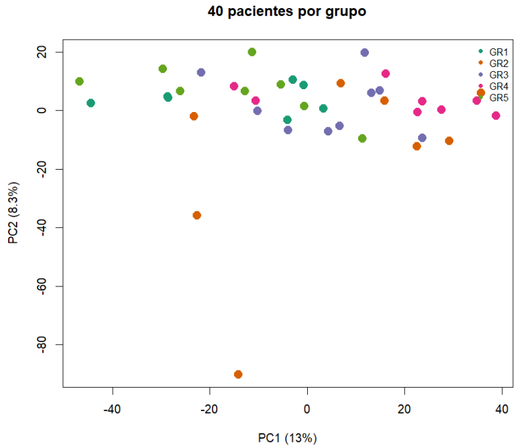


Neg 2

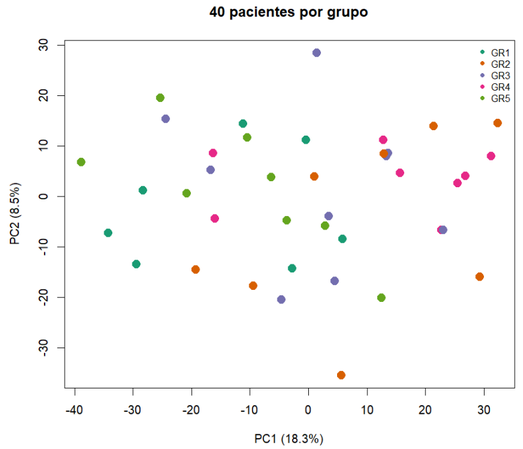
PCA de los 40 pacientes



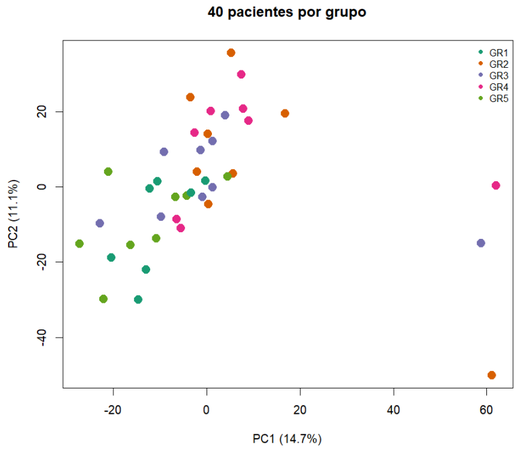
Pos 1



Pos 2

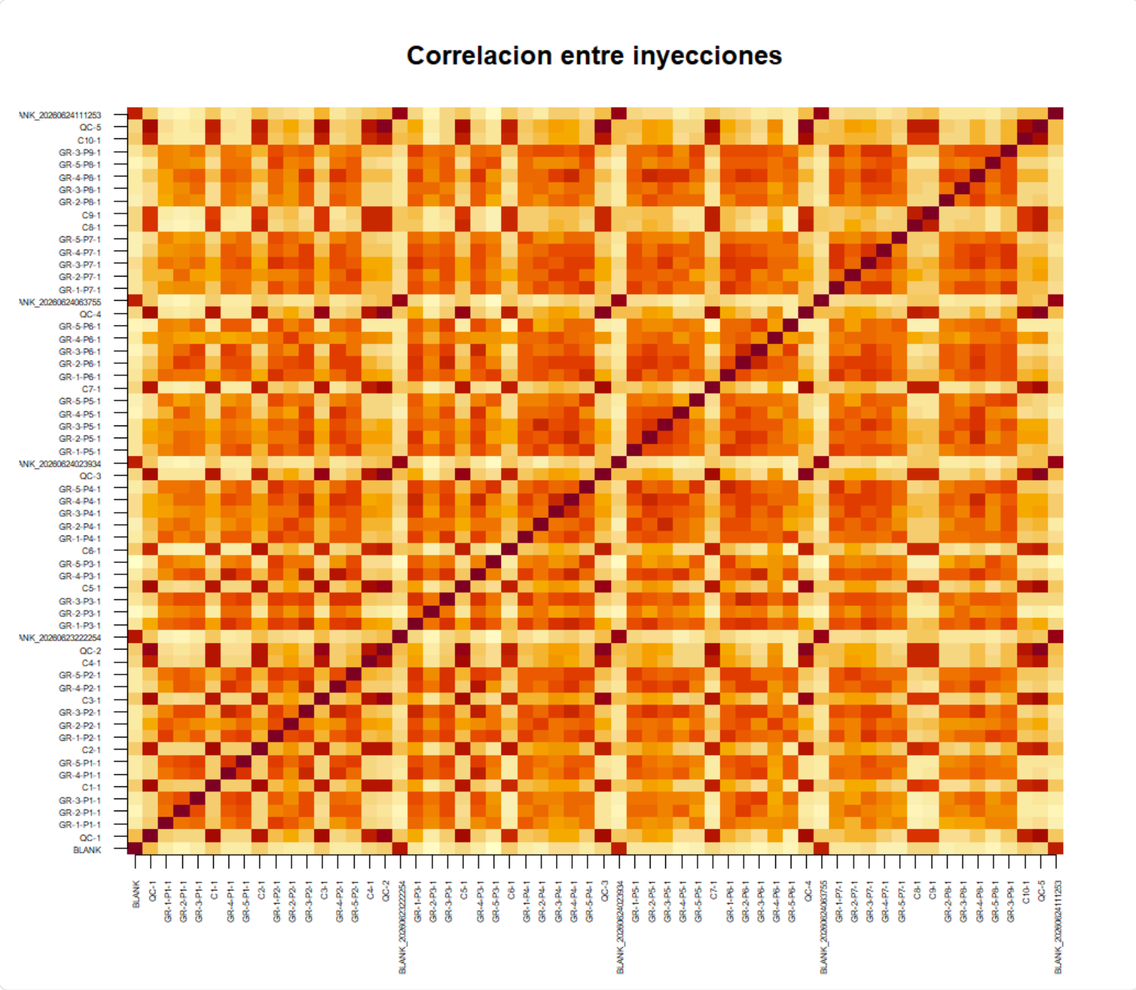


Neg 1

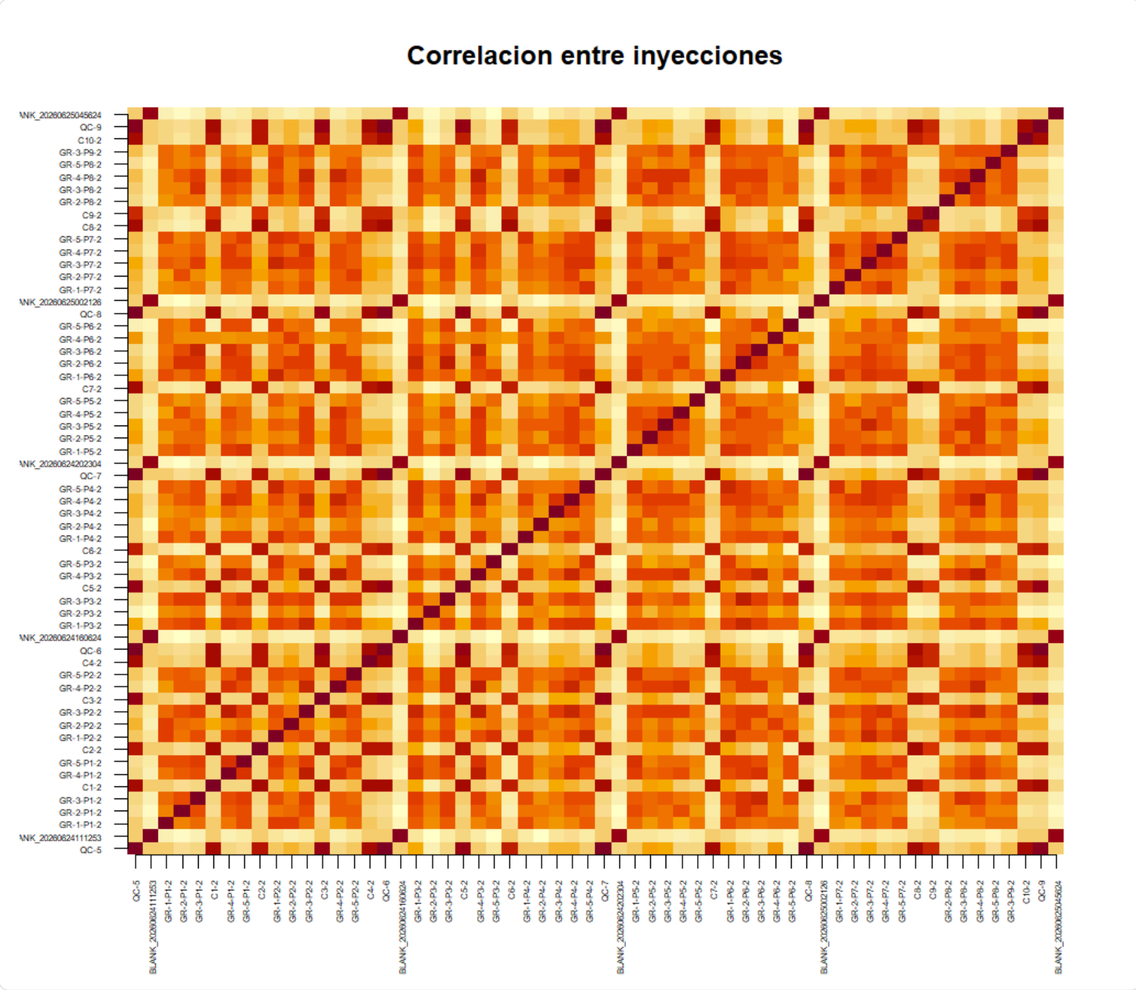


Neg 2

Correlación entre todas las inyecciones, positivo

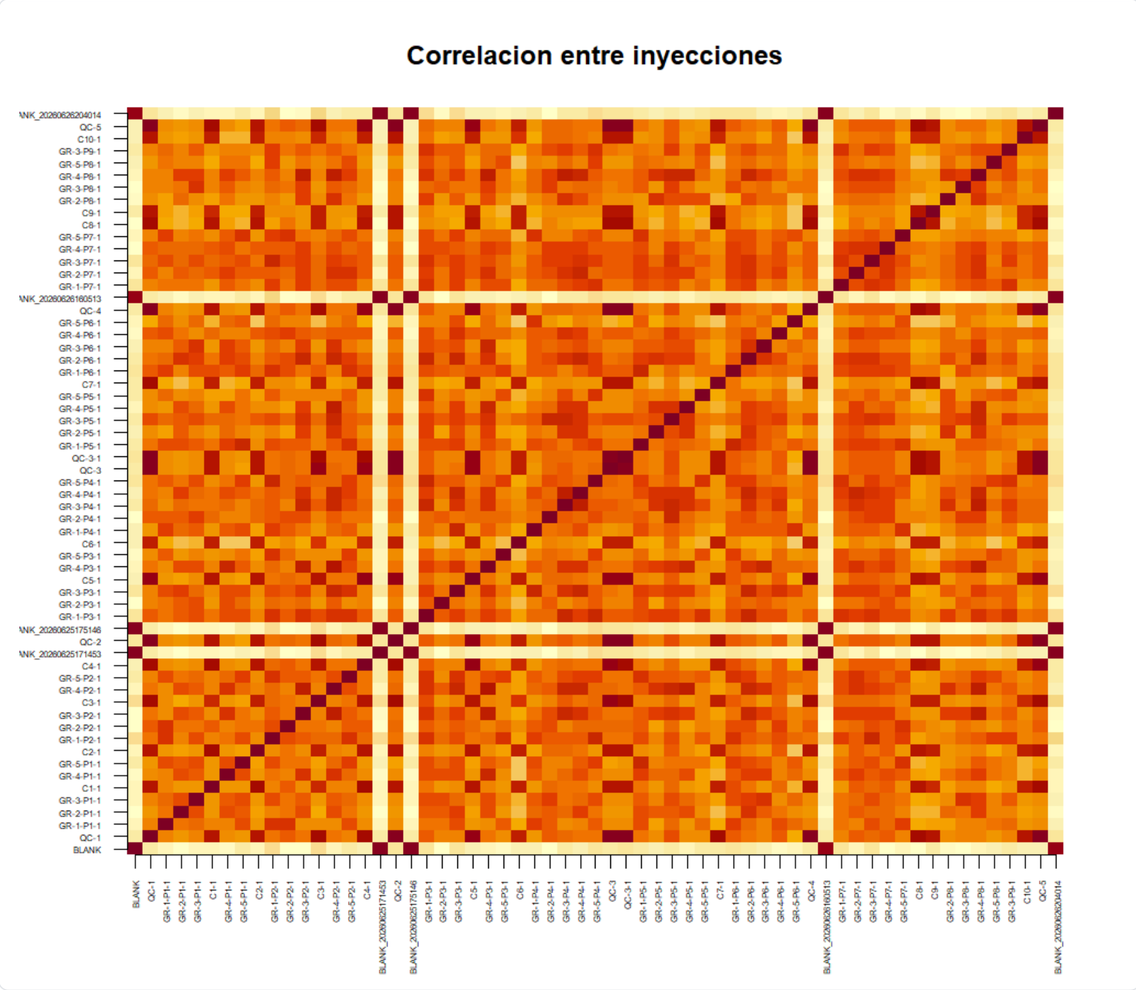


Tanda 1

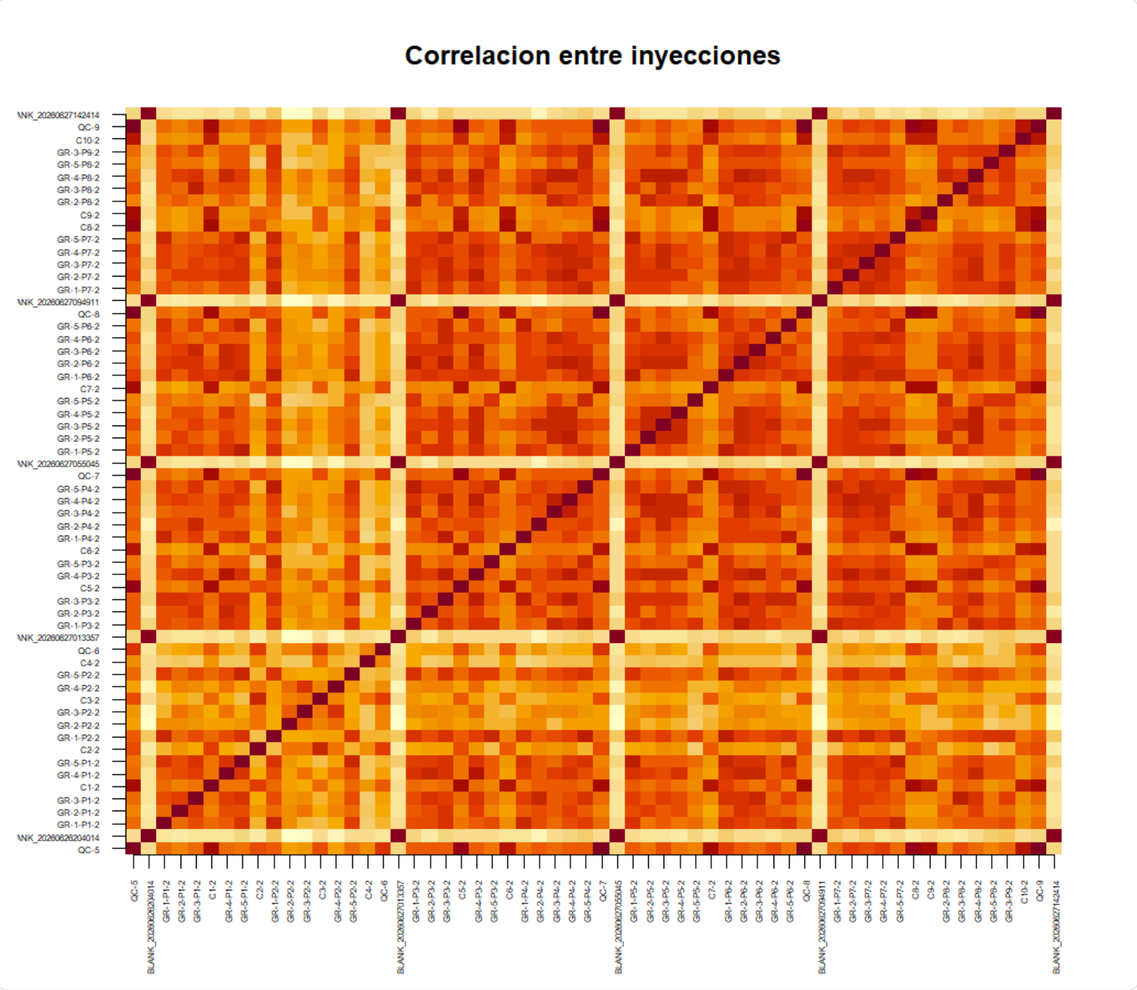


Tanda 2

Correlación entre todas las inyecciones, negativo

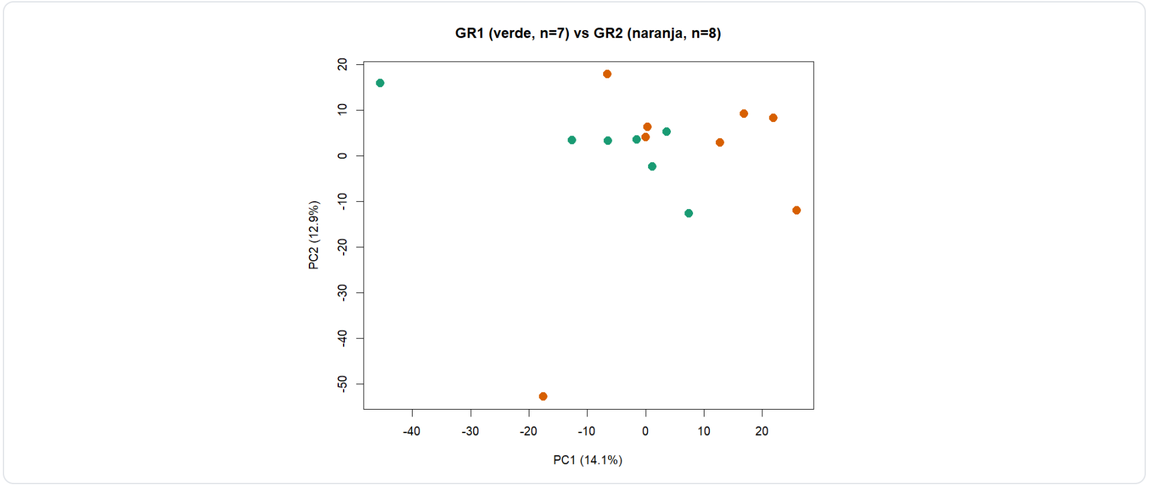


Tanda 1

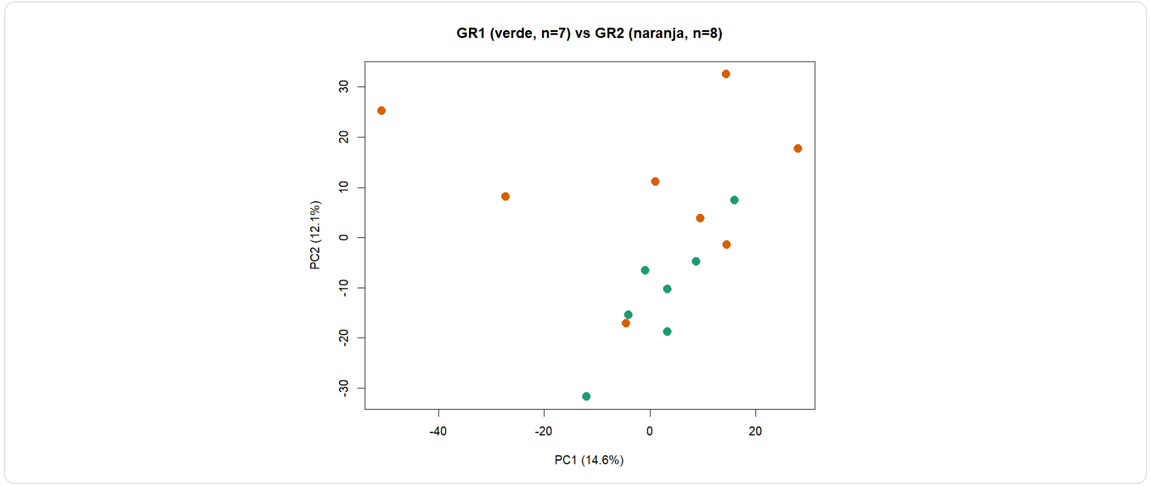


Tanda 2

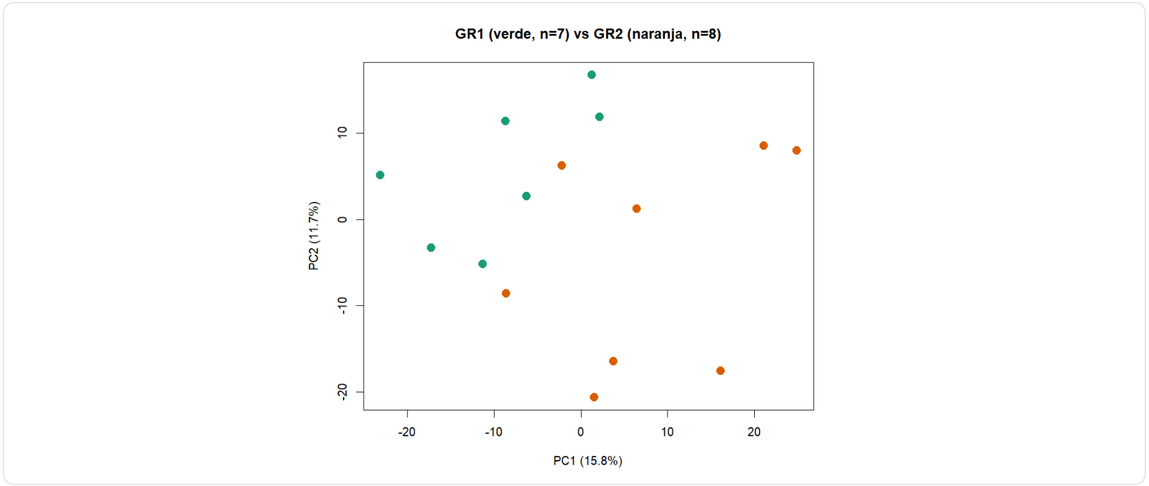
PCA de GR1 frente a GR2



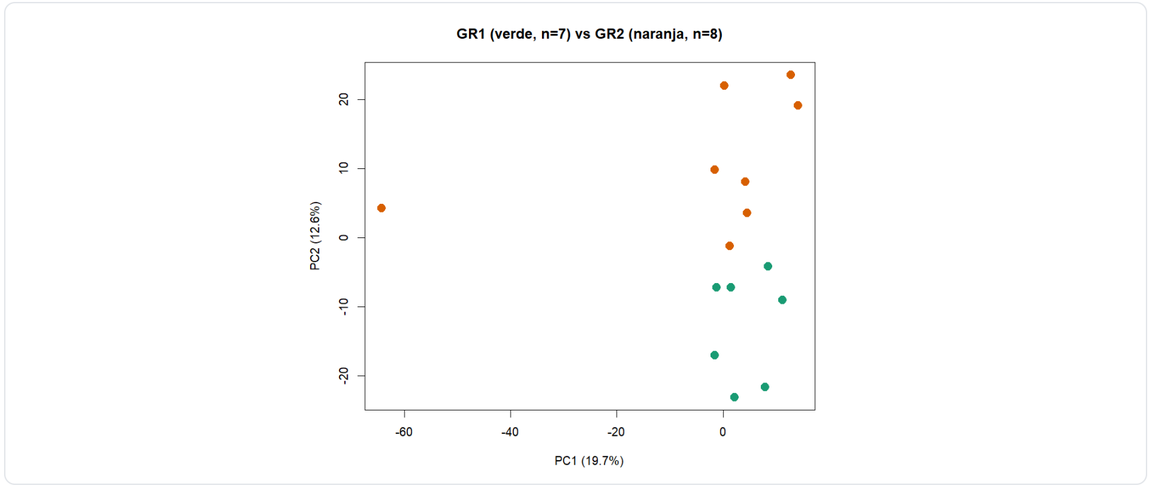
Pos 1



Pos 2



Neg 1



Neg 2

GR1 frente a GR2 en negativo, tanda a tanda

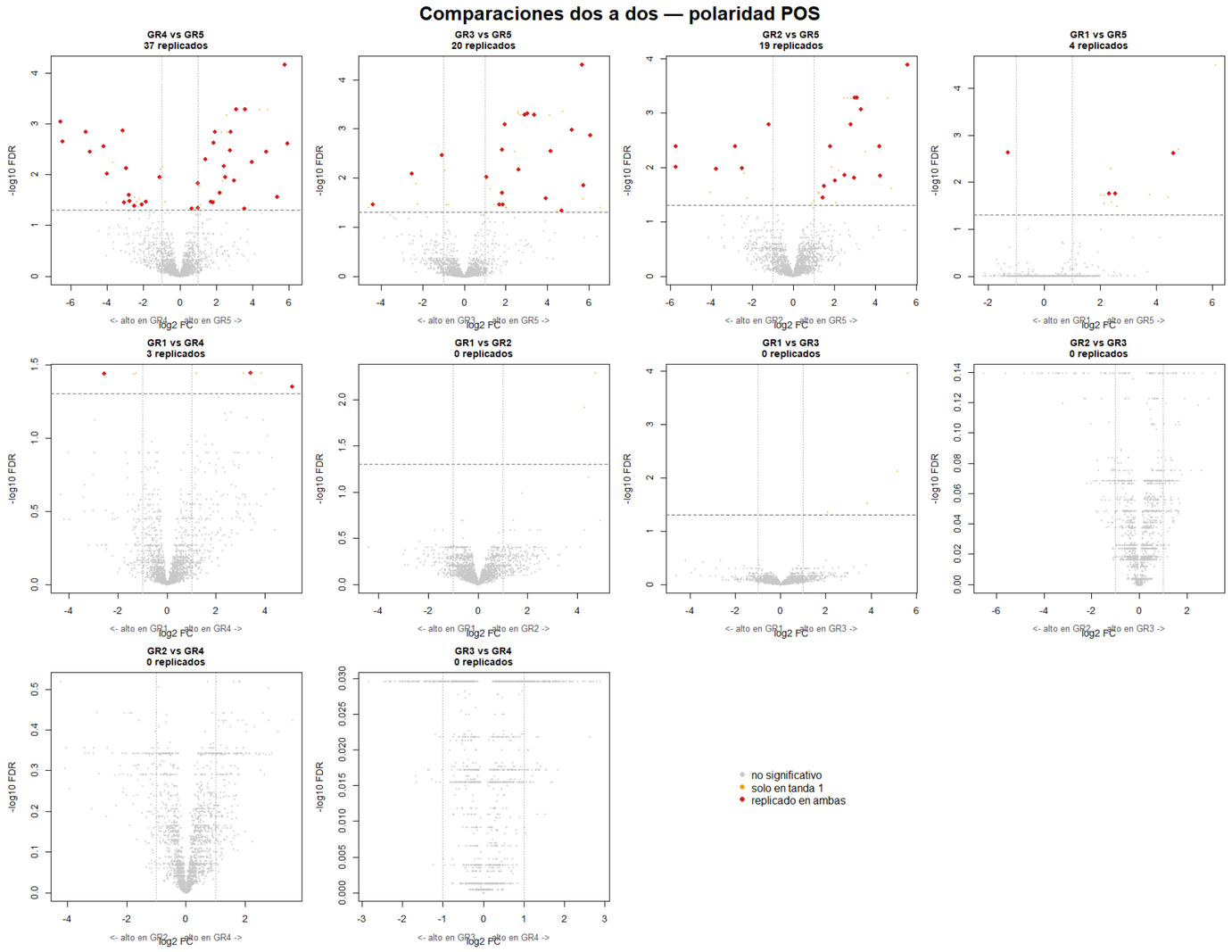
Tanda 1; 1.151 testados

m/z	RT	veces	FDR
204,9997	3,64	12,8	0,0068
246,9948	3,55	15,0	0,0098
133,0662	4,66	28,9	0,0153
203,0038	3,63	8,5	0,0209
367,1087	4,06	28,2	0,0209
217,0198	4,08	15,2	0,0287
153,0201	3,91	2,9	0,0421
397,2321	11,86	2,6	0,0421
139,0769	3,77	2,4	0,0421

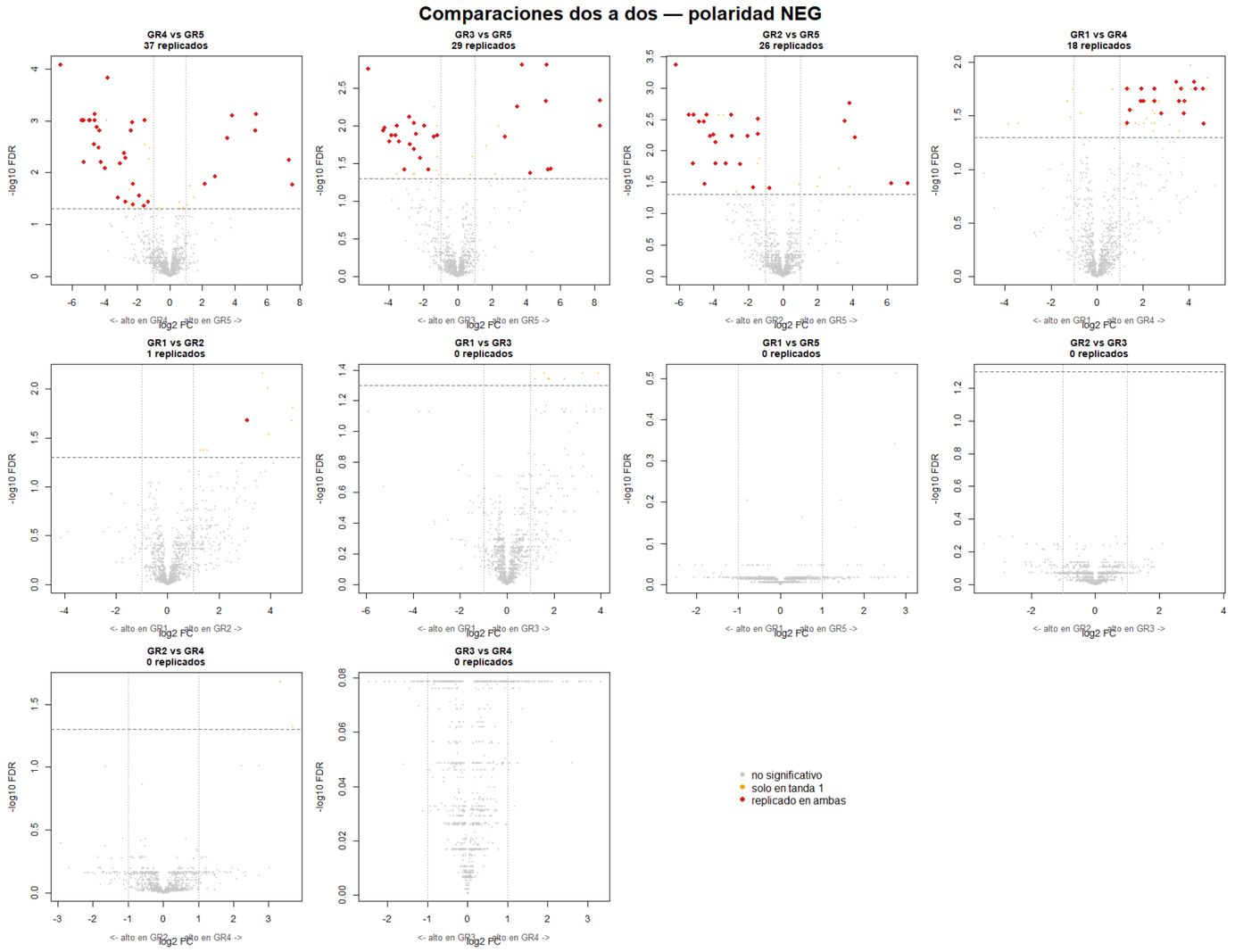
Tanda 2; 1.747 testados

m/z	RT	veces	FDR
295,9613	3,80	29,7	0,0149
204,0073	3,65	7,2	0,0149
123,0453	3,68	3,7	0,0189
245,0155	3,93	20,9	0,0237
273,0112	3,98	13,5	0,0303
554,3100	7,88	22,0	0,0347
203,0038	3,63	6,0	0,0496

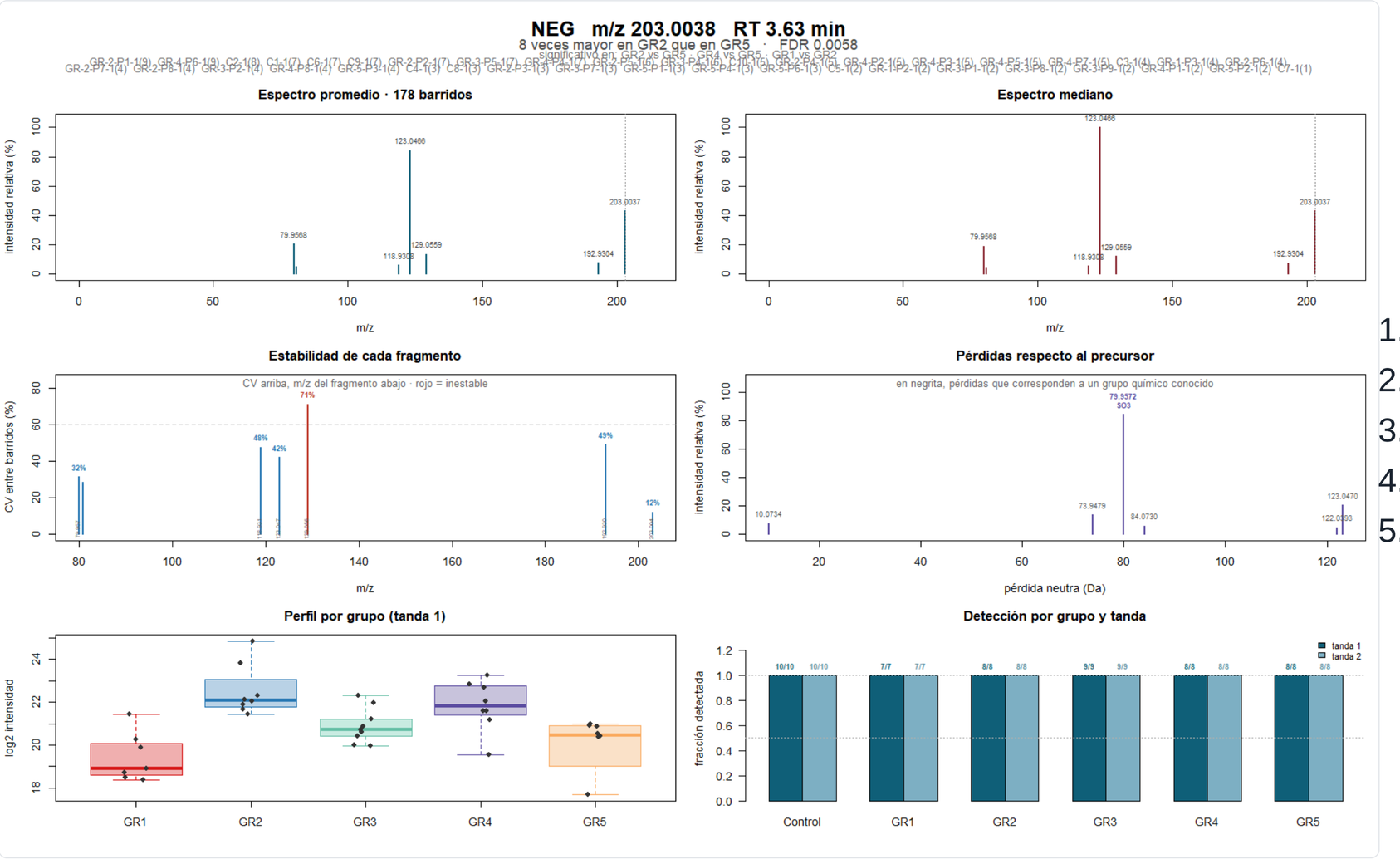
Volcanos de las diez parejas, positivo



Volcanos de las diez parejas, negativo

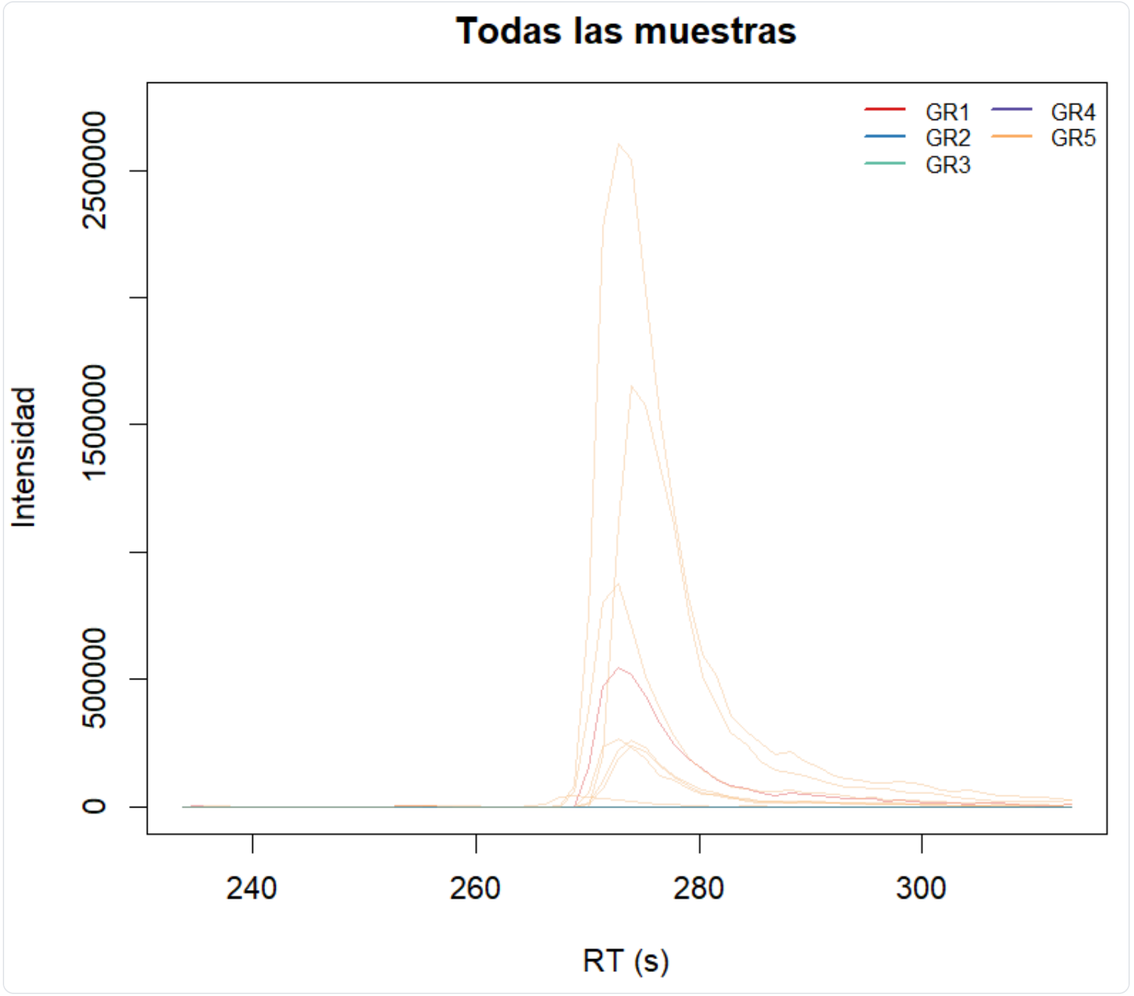


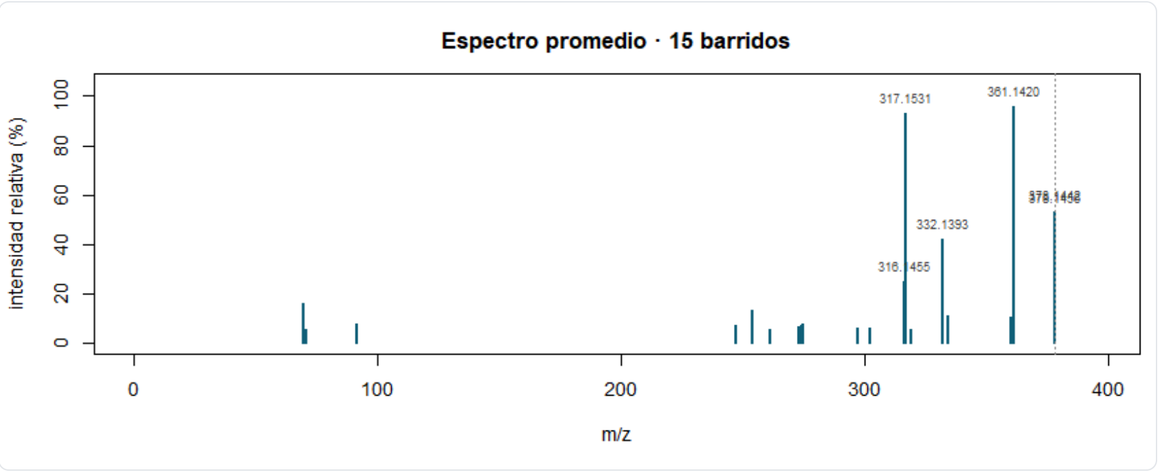
Fragmentación del 203,0038

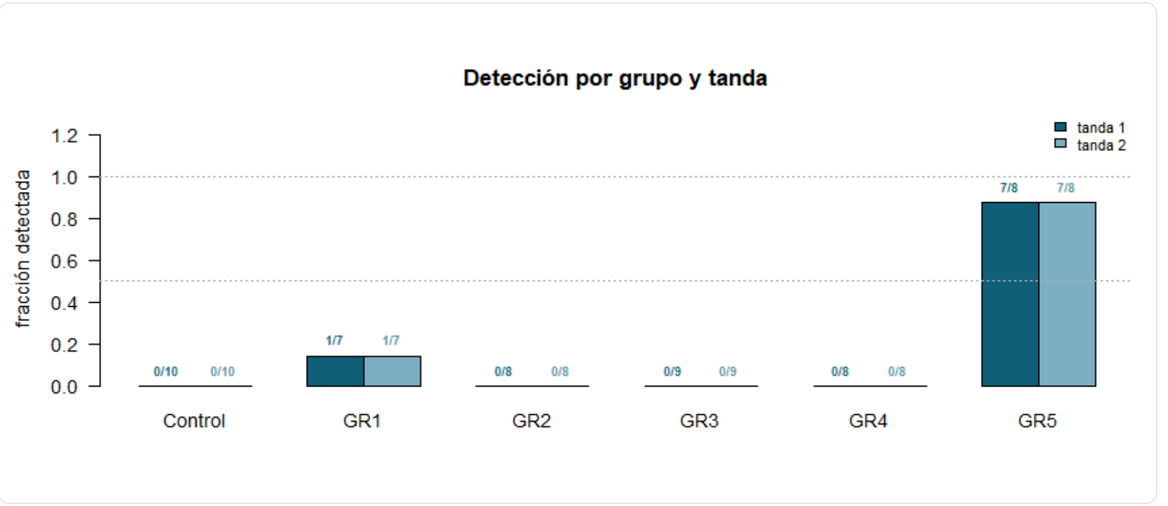


1. Espectro promedio y mediano
2. Estabilidad de cada fragmento
3. Pérdidas respecto al precursor
4. Perfil por grupo
5. Detección por grupo y tanda

Levofloxacino

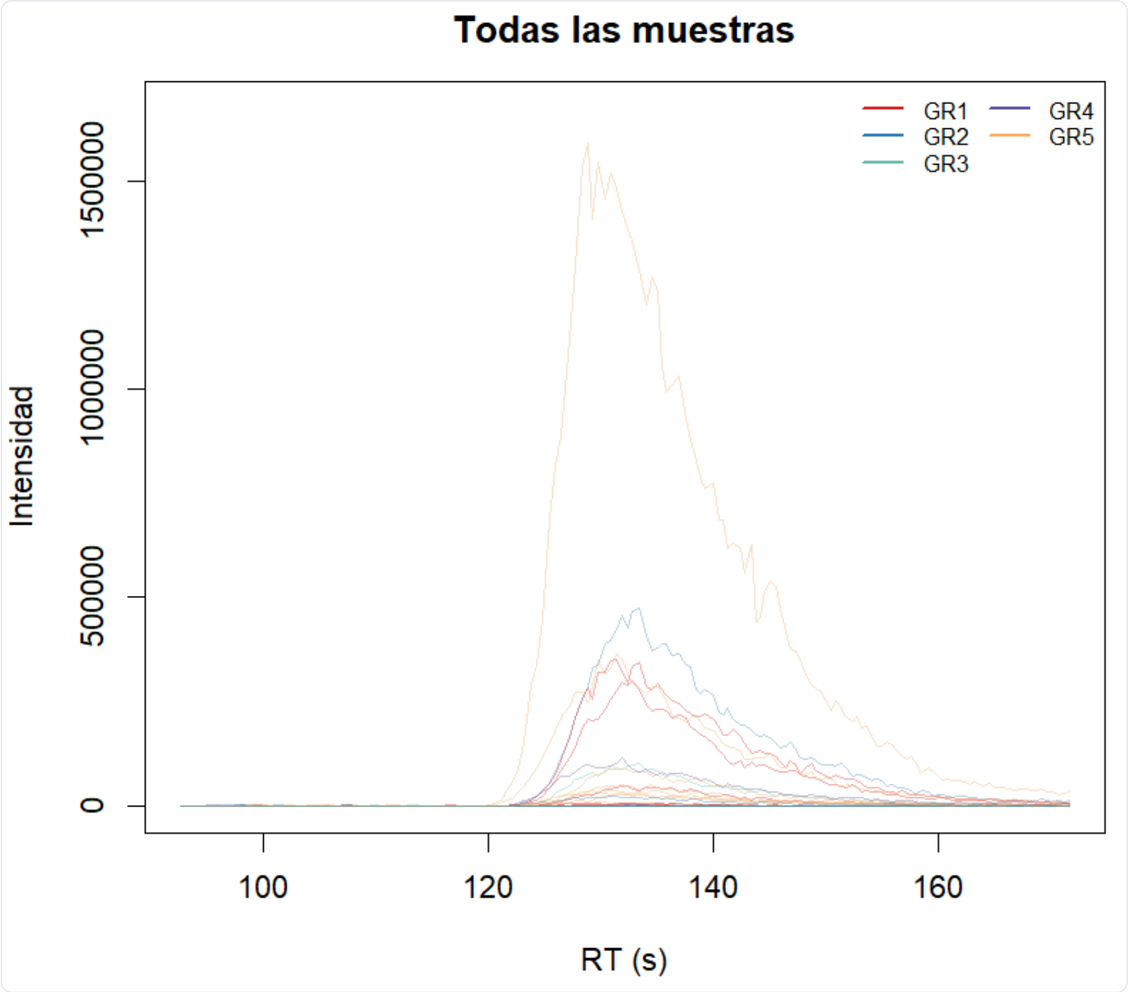


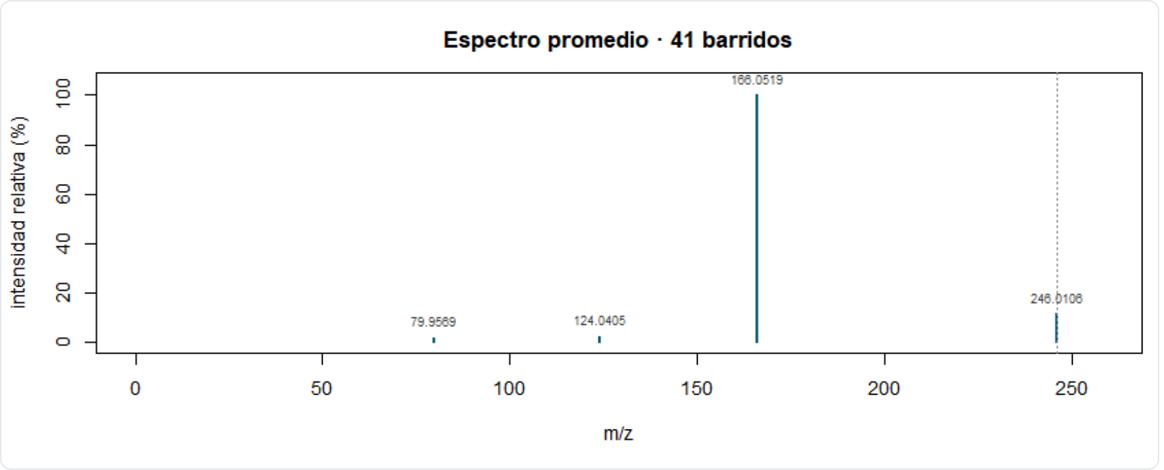


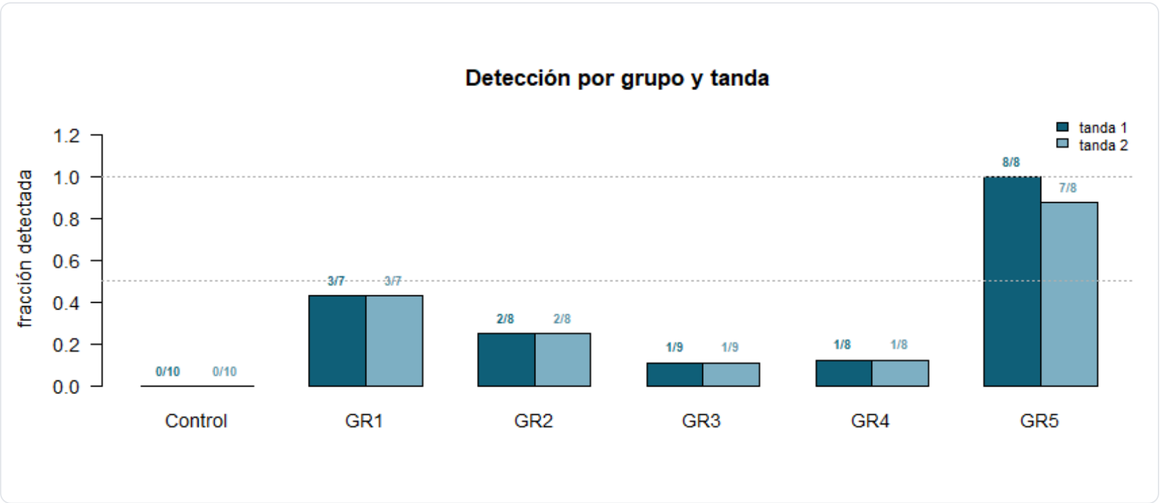


m/z 378,146; RT 4,56 min; ESI+

Paracetamol

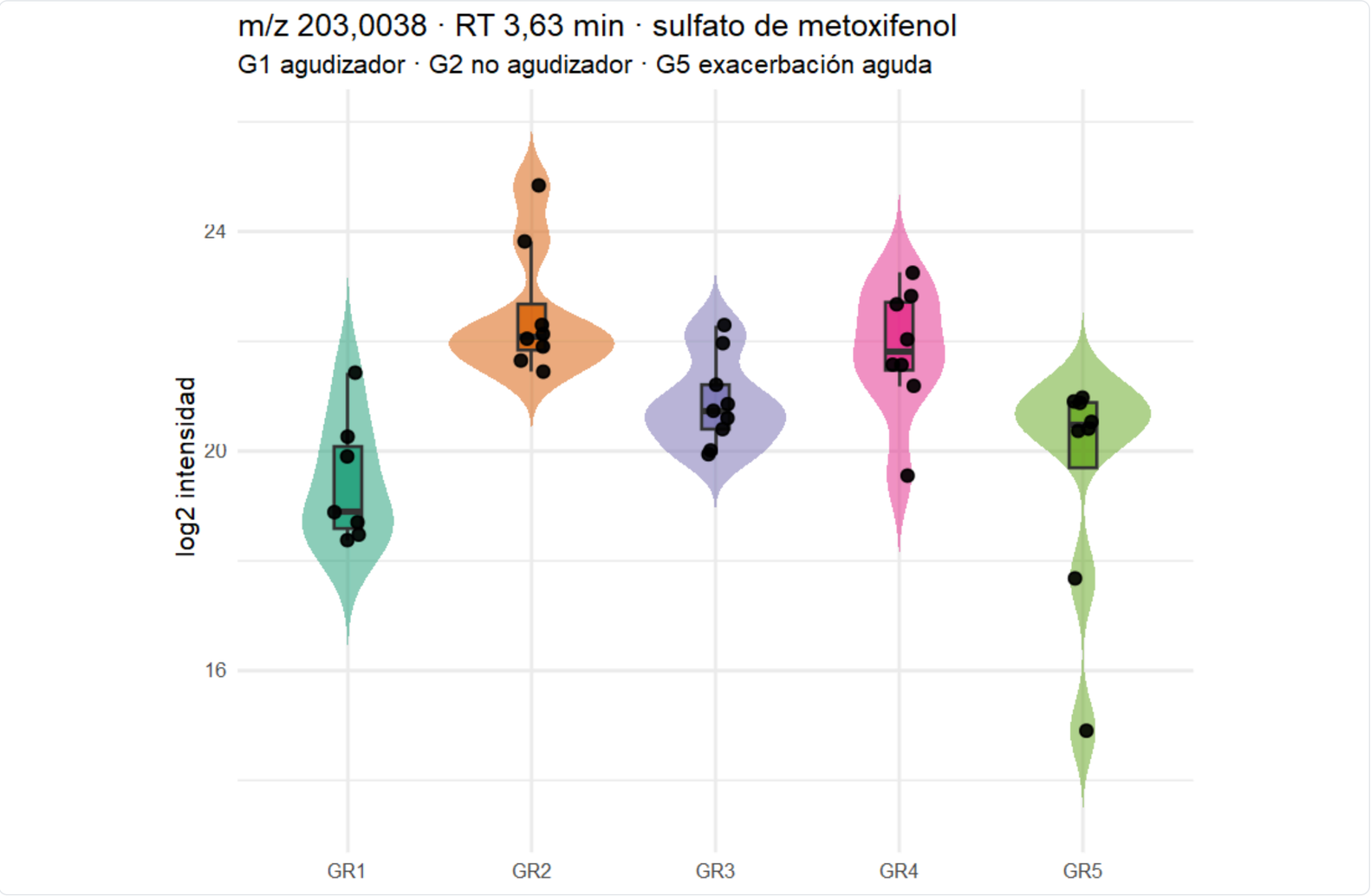






m/z 246,011; RT 2,20 min; ESI-

Sulfato de metoxifenol en los cinco grupos

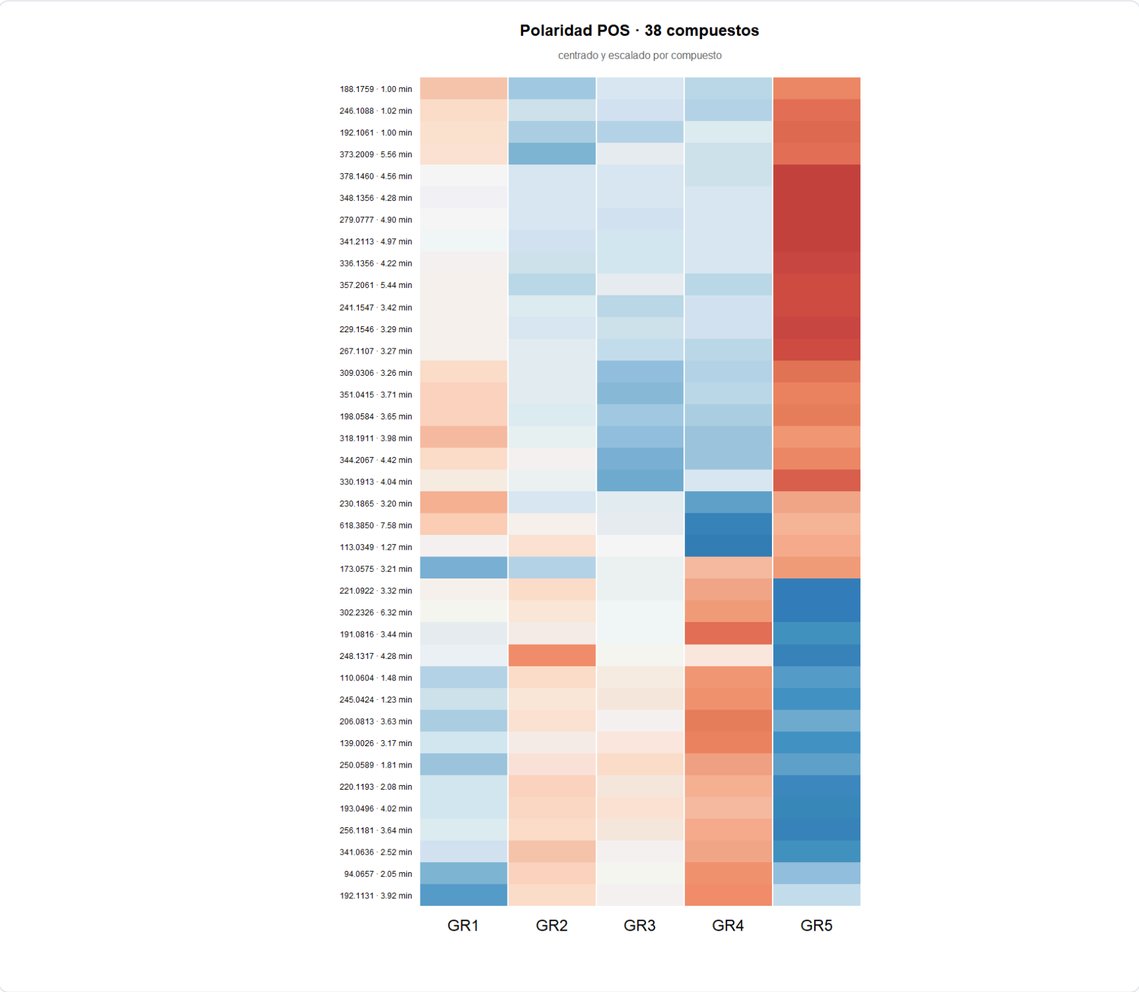


Bajo en los que agudizan, GR1 y GR5

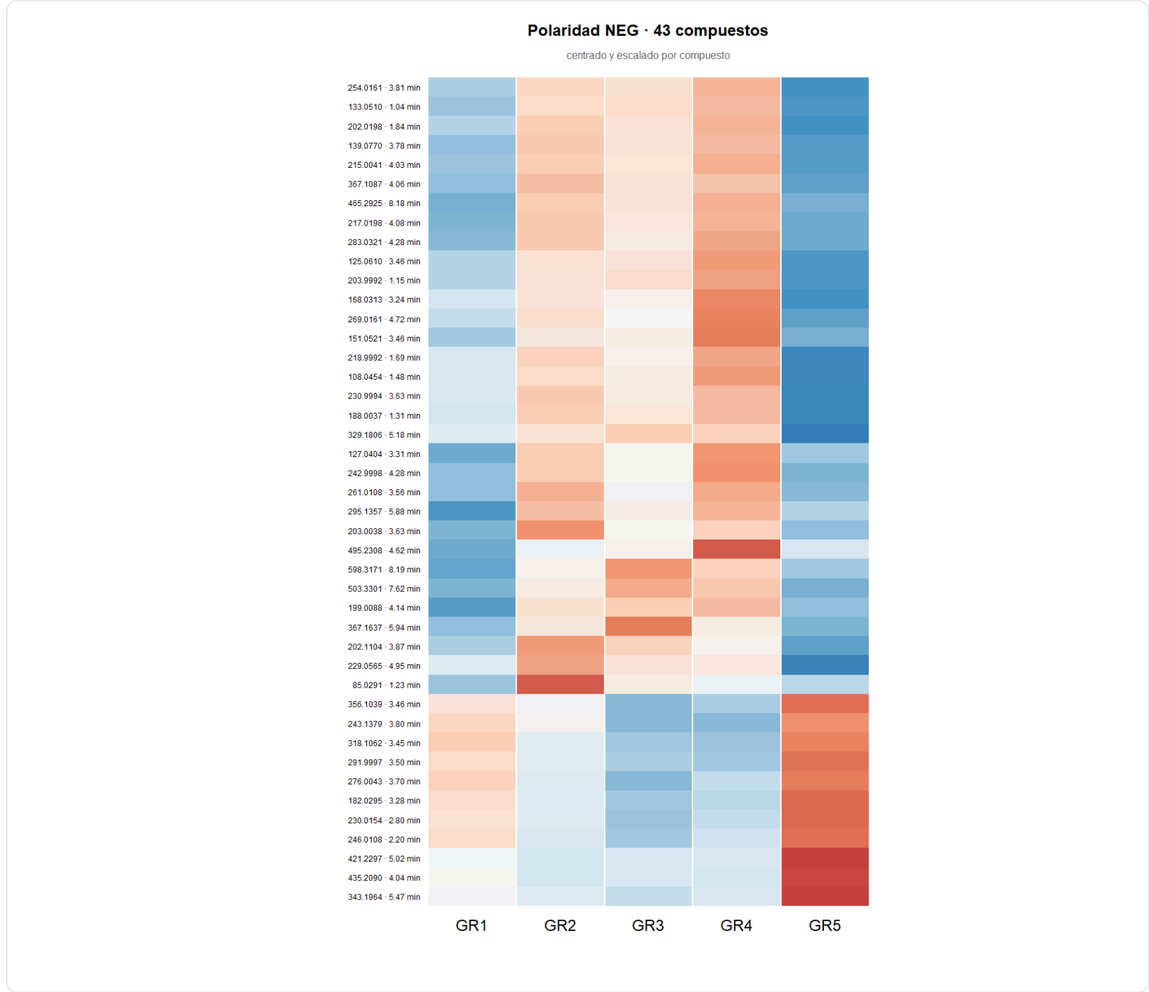
Este perfil hizo pasar de dos grupos a los cinco.

m/z 203,0038; RT 3,63 min; ESI-

Mapas de calor de los cinco grupos



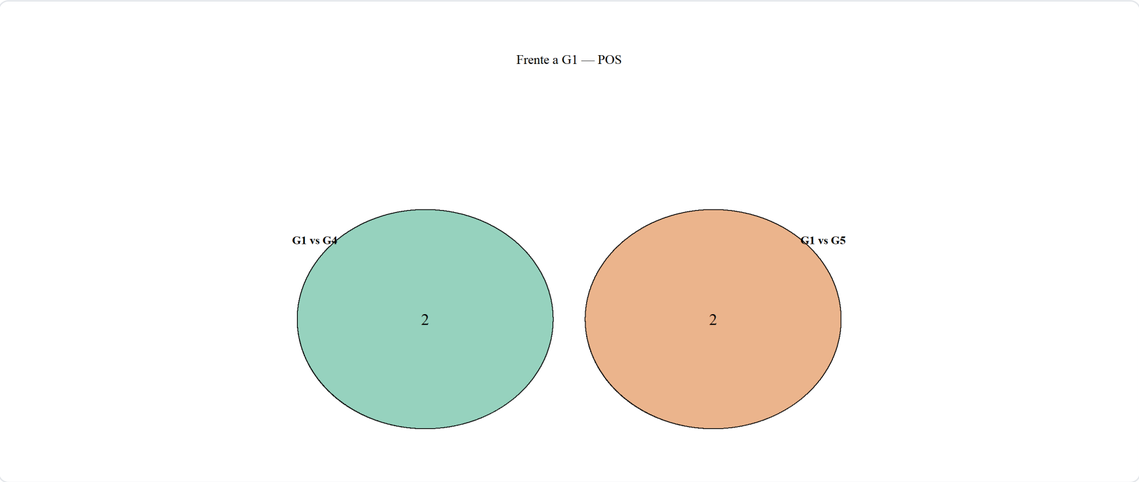
Positivo, 38



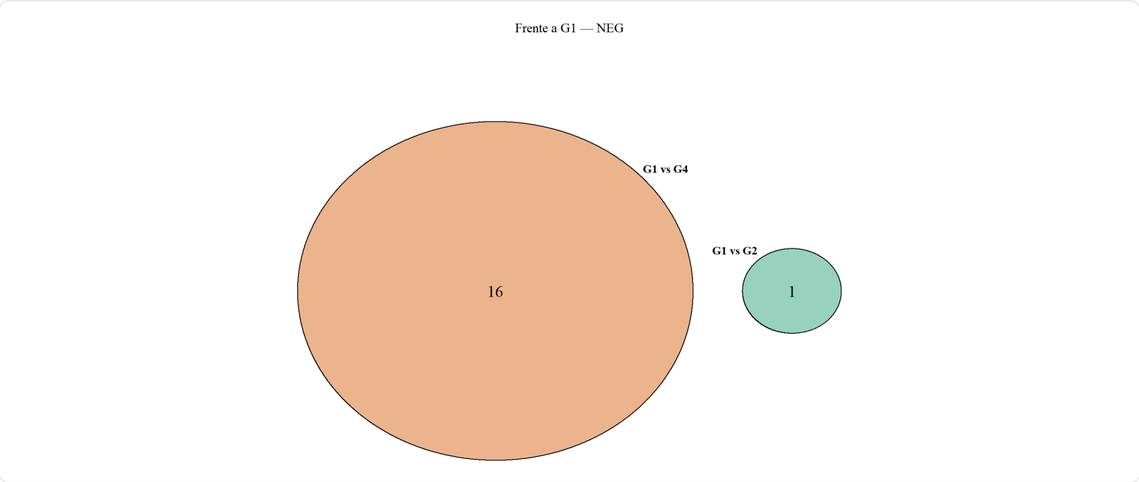
Negativo, 43

FDR < 0,05 en ambas tandas

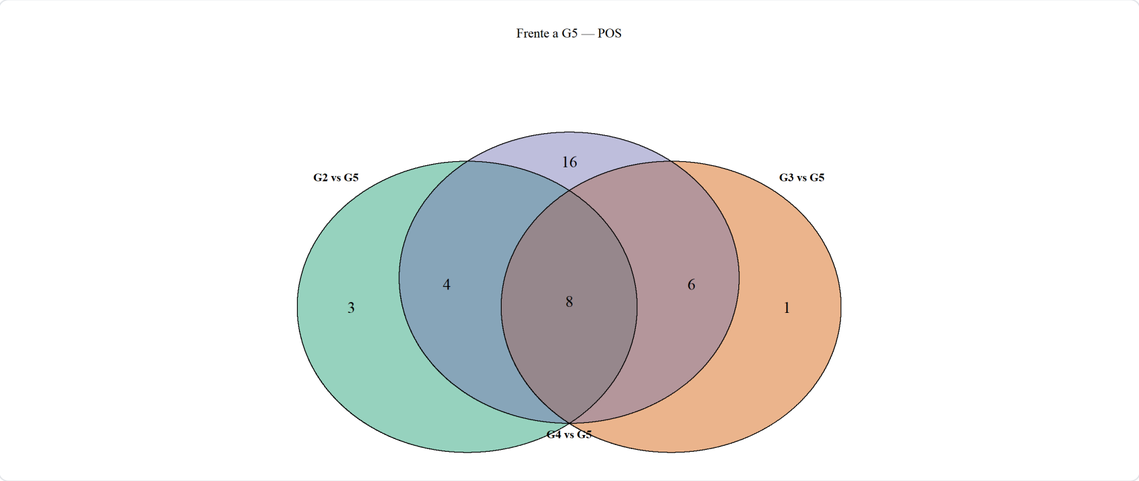
Diagramas de Venn



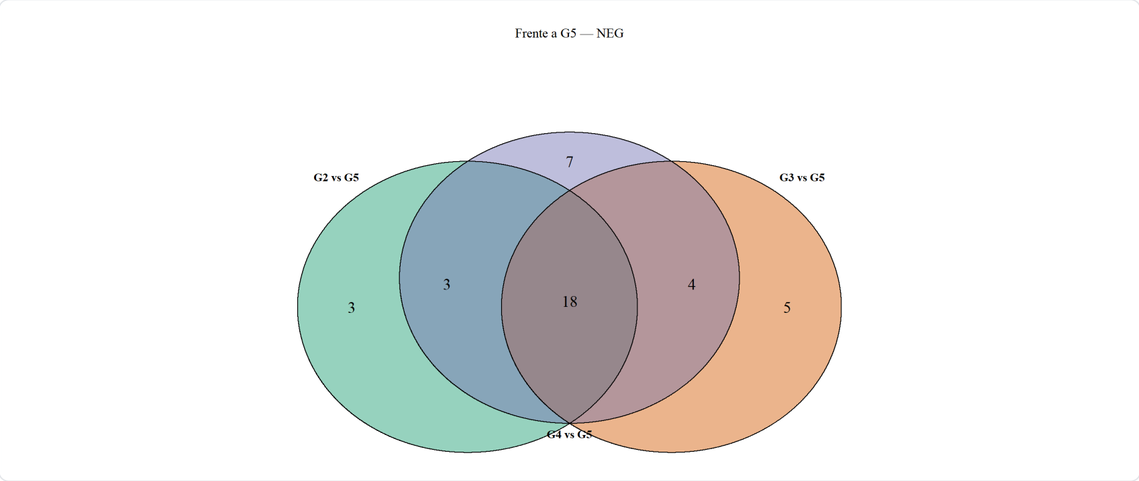
Frente a GR1, positivo



Frente a GR1, negativo

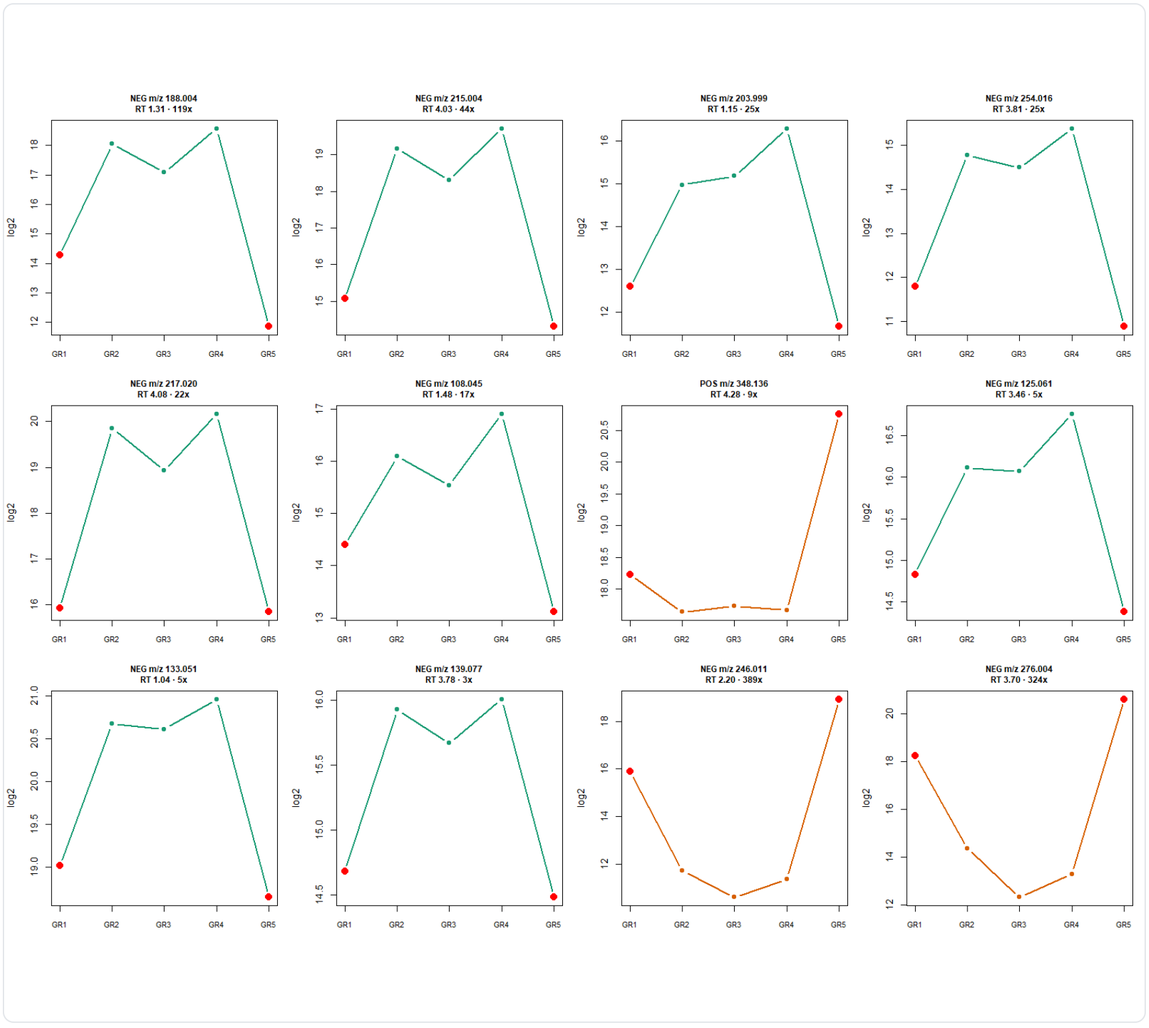


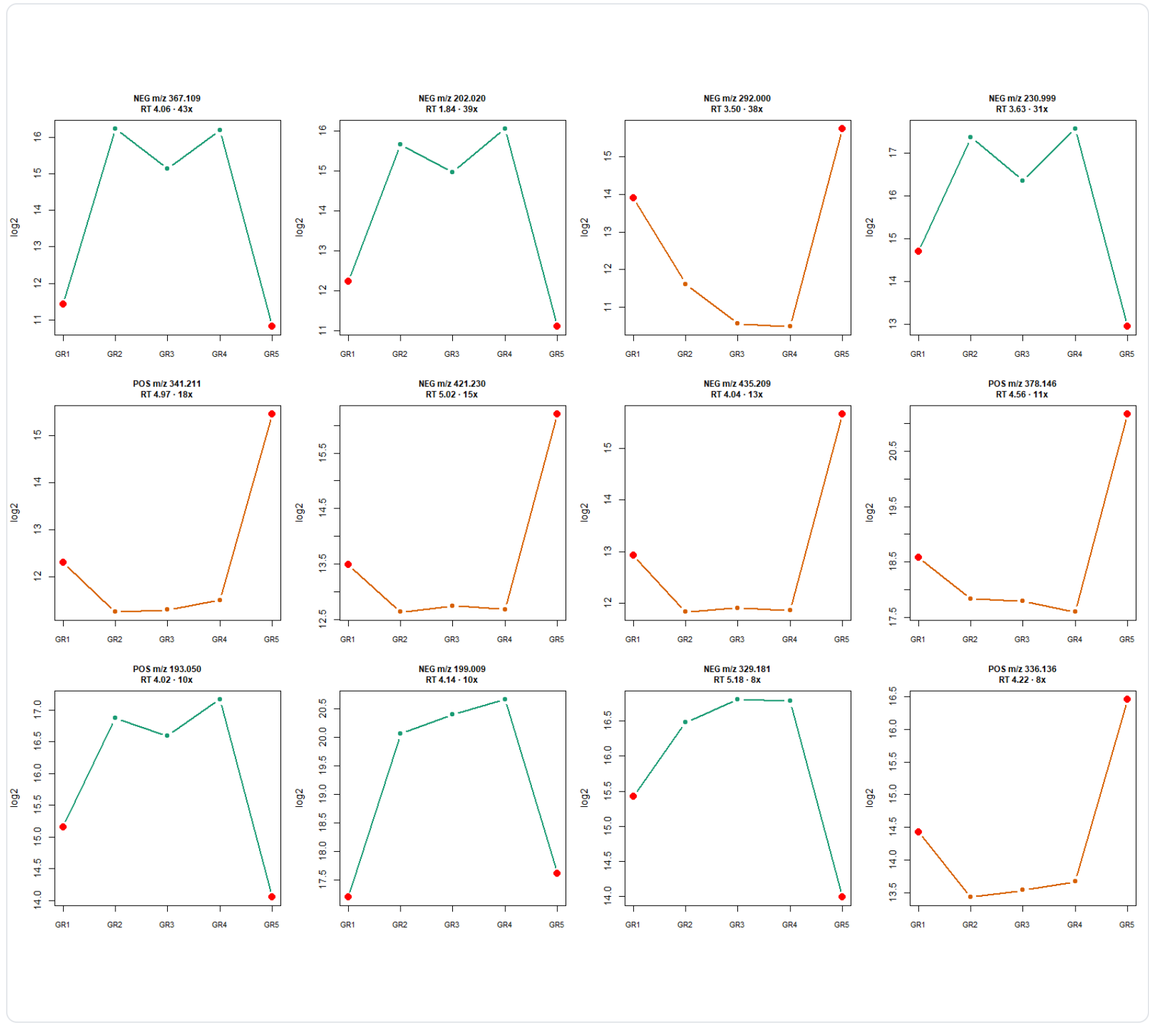
Frente a GR5, positivo, 8 en las tres



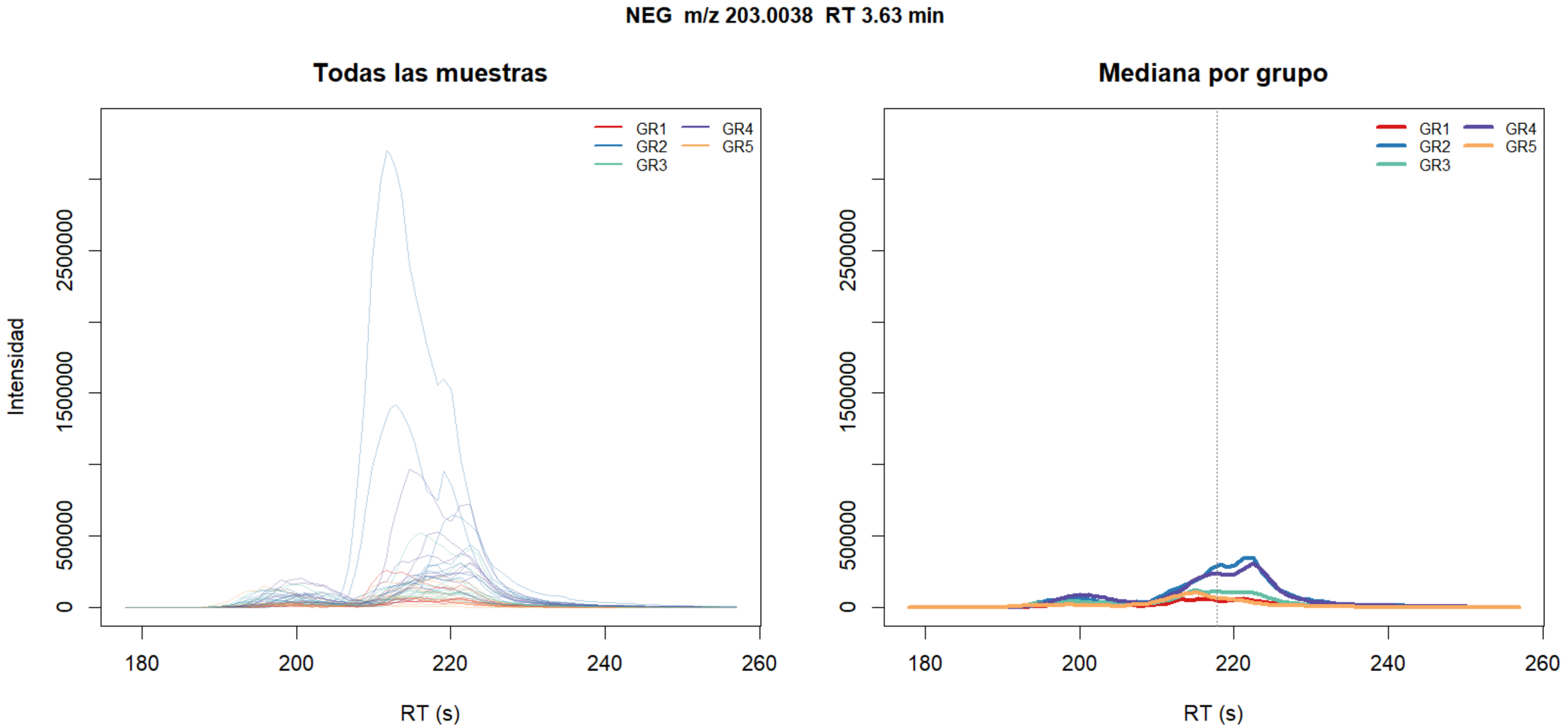
Frente a GR5, negativo, 18 en las tres

Perfiles de GR1 a GR5



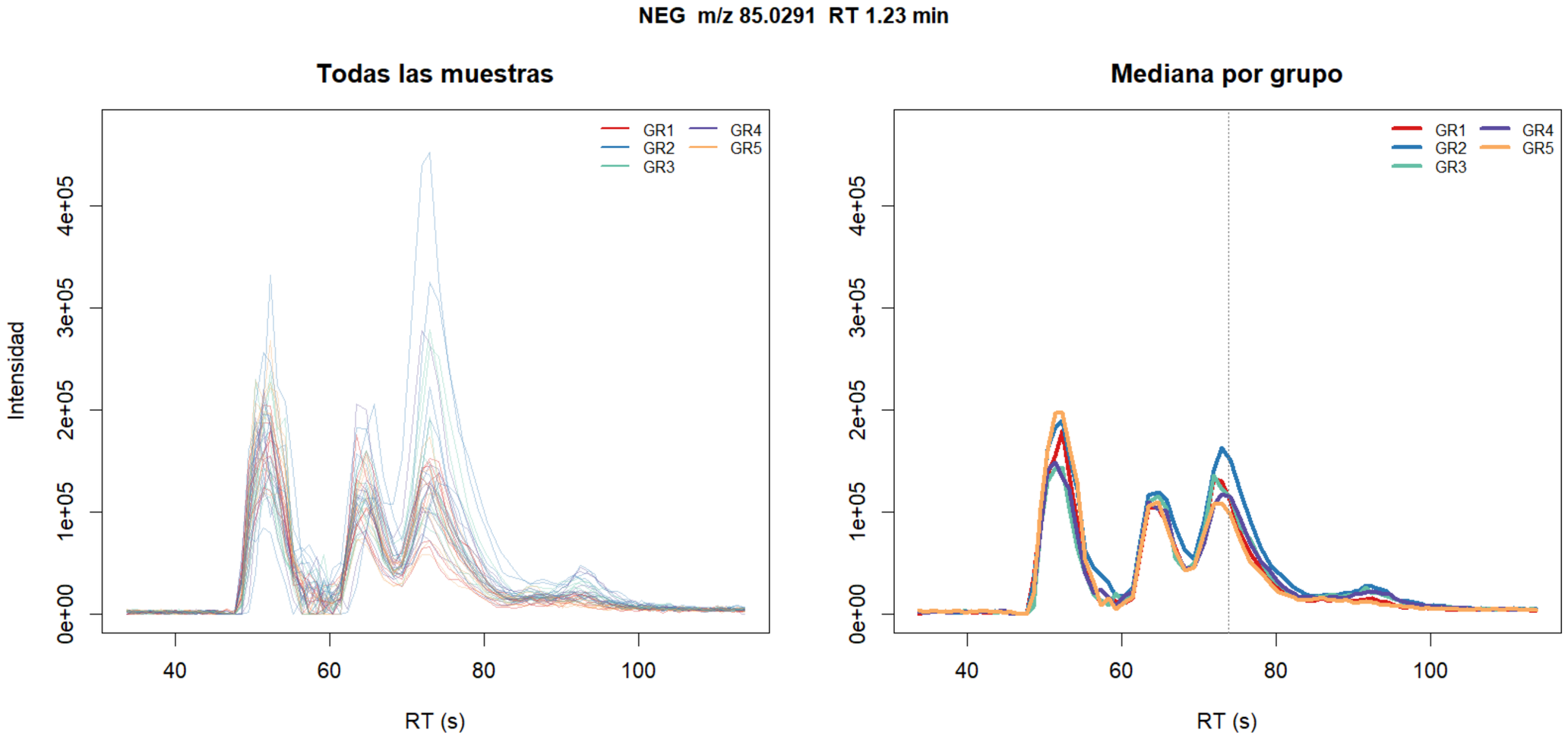


EIC del 203,0038



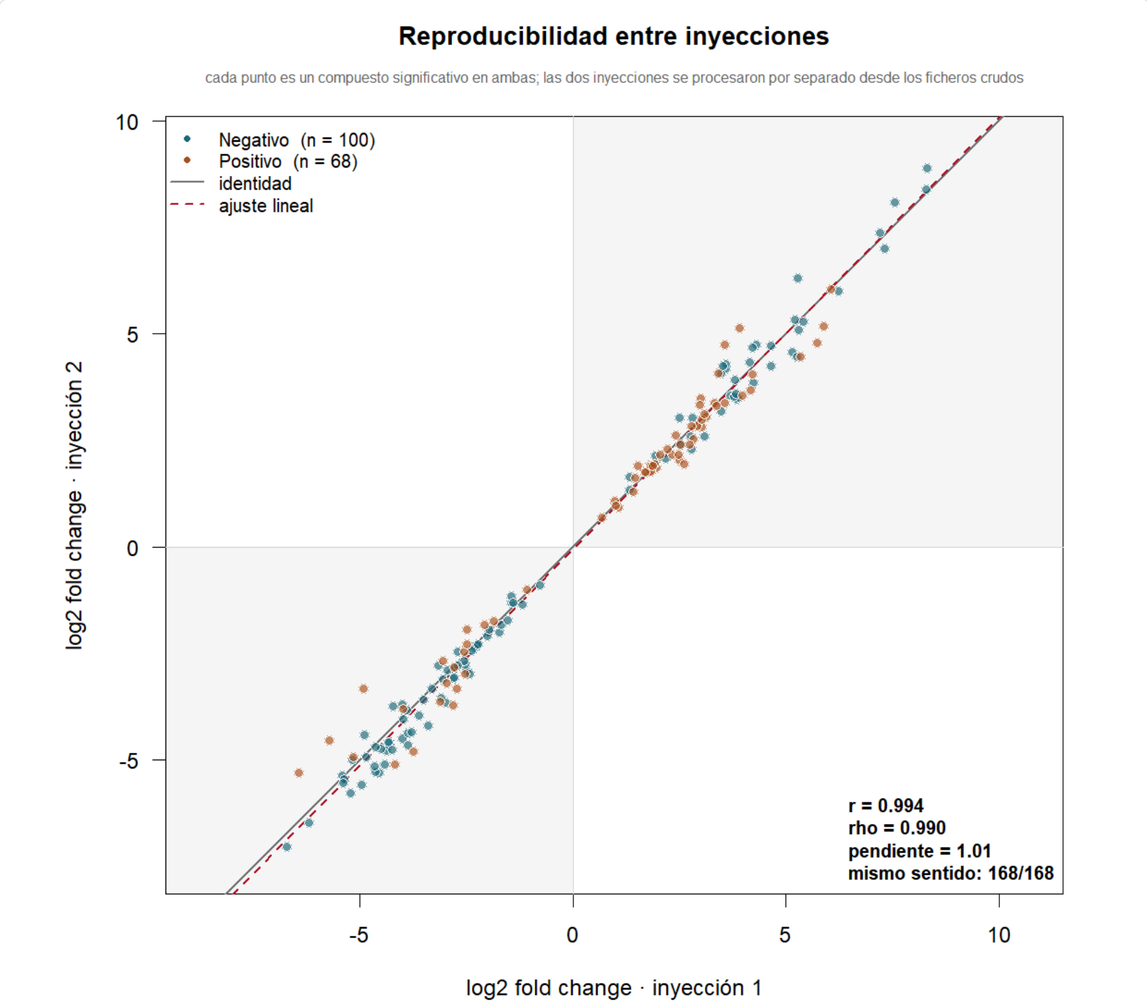
m/z 203,0038; RT 3,63 min

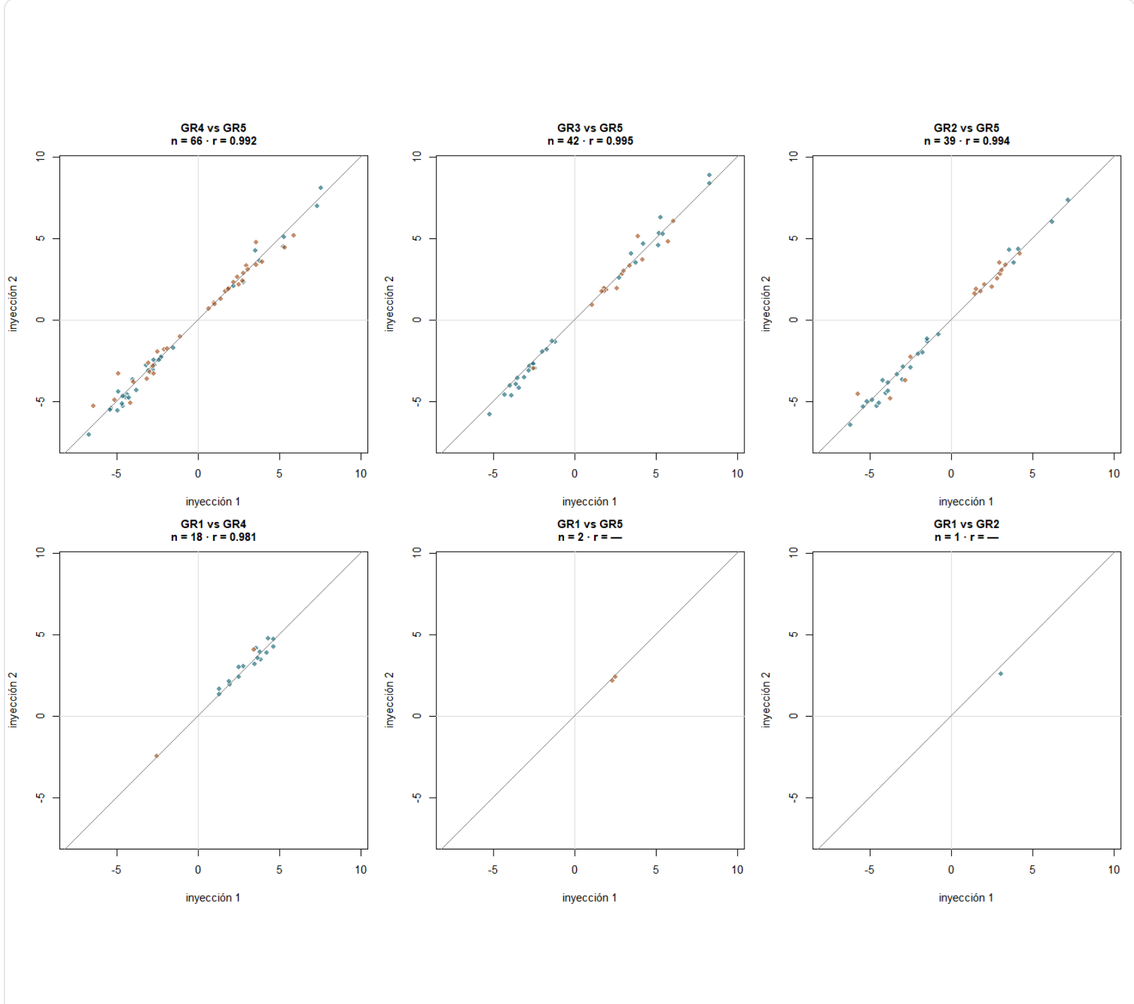
Compuesto descartado



m/z 85,0291; RT 1,23 min

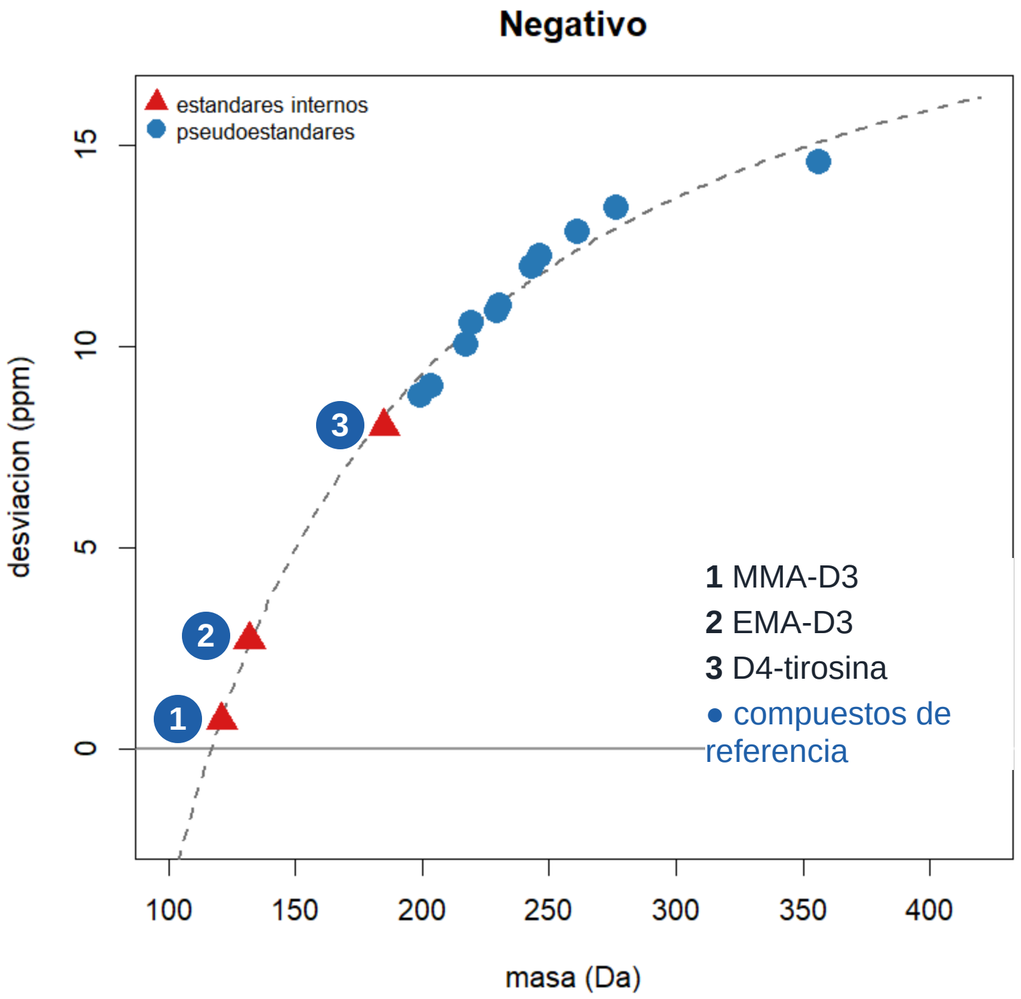
Inyección 1 frente a inyección 2

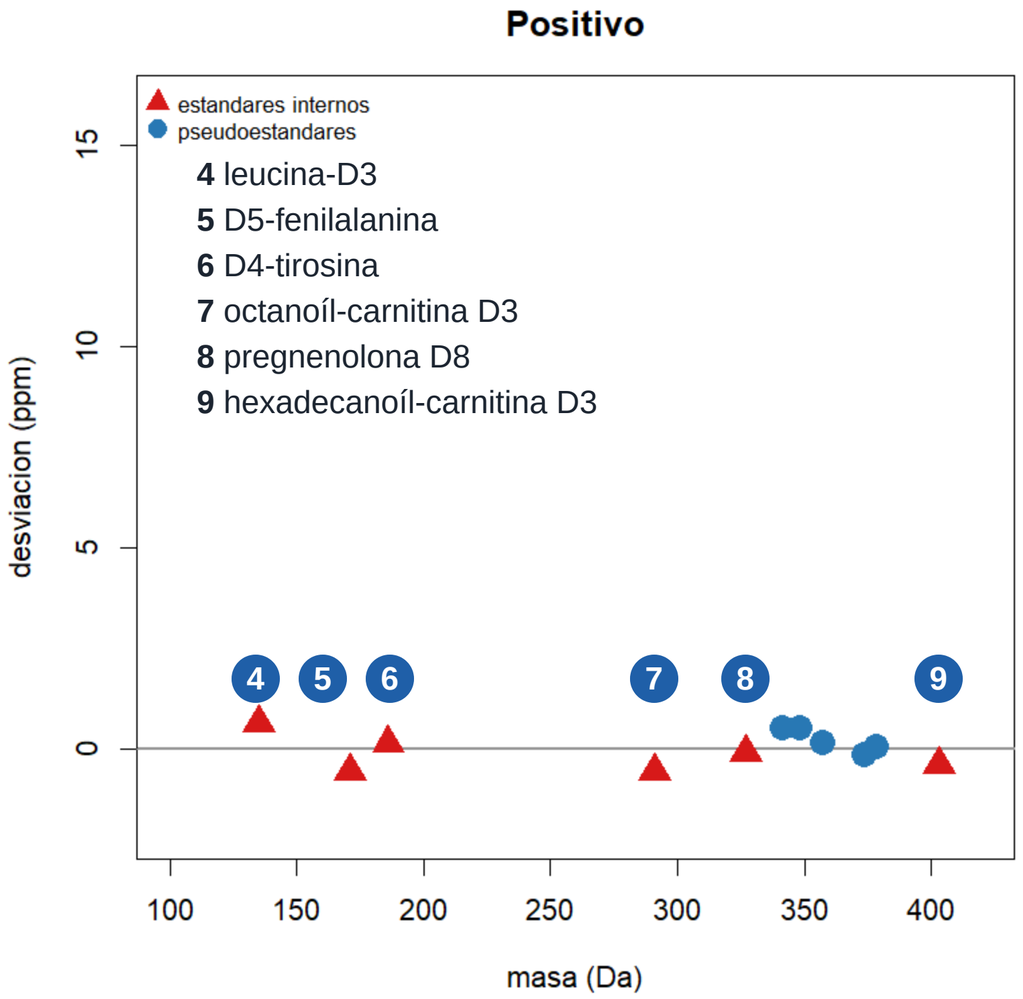




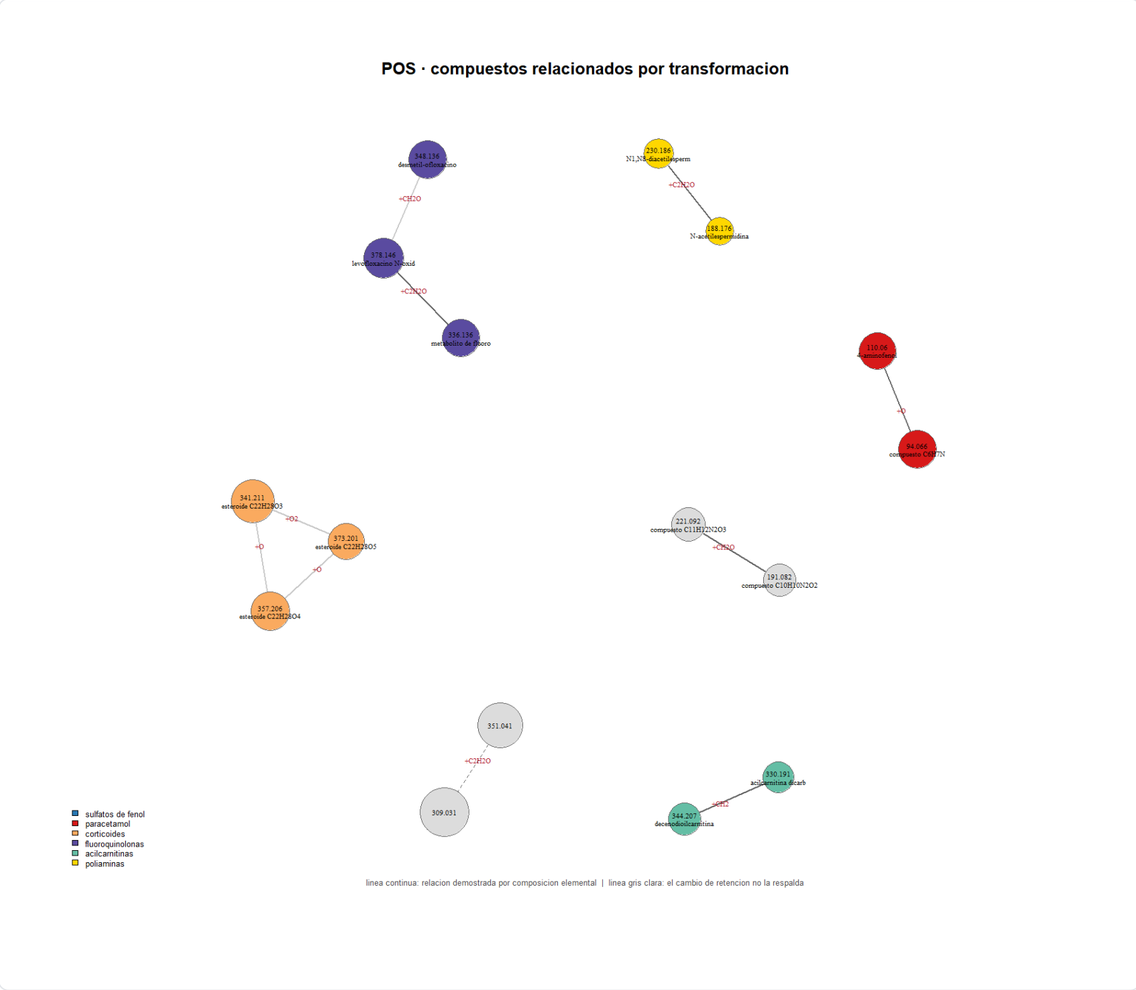
r = 0,994; pendiente 1,01

Desviación de masa antes de corregir

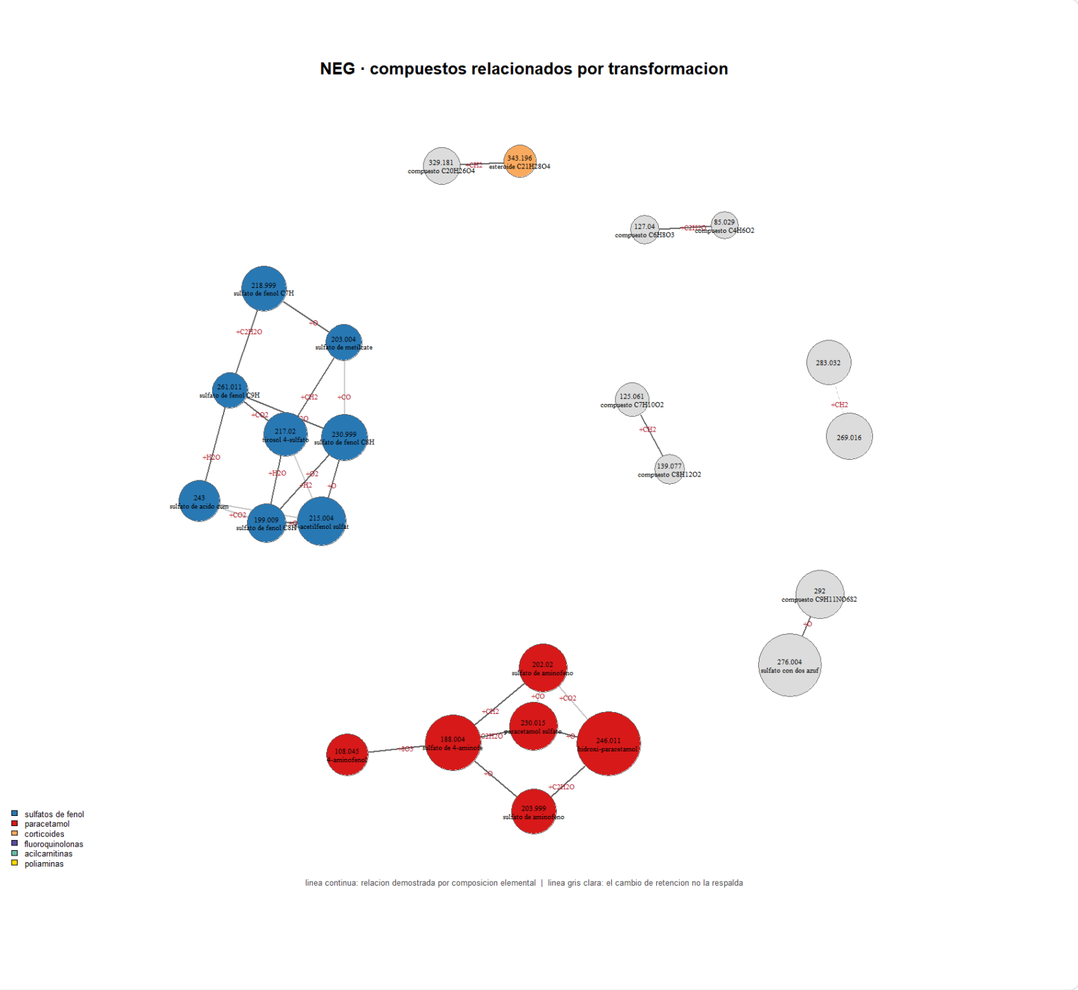




Red de transformaciones

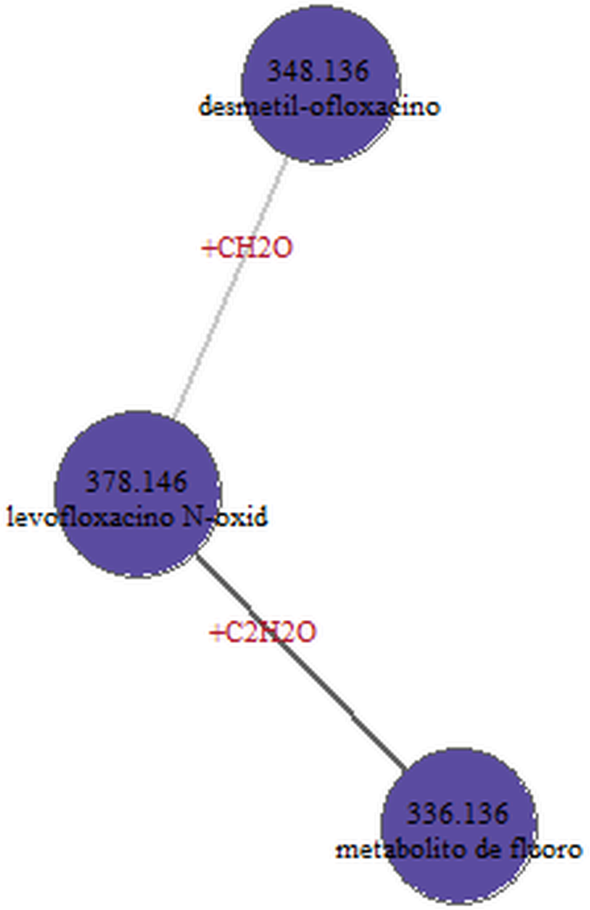


Positivo

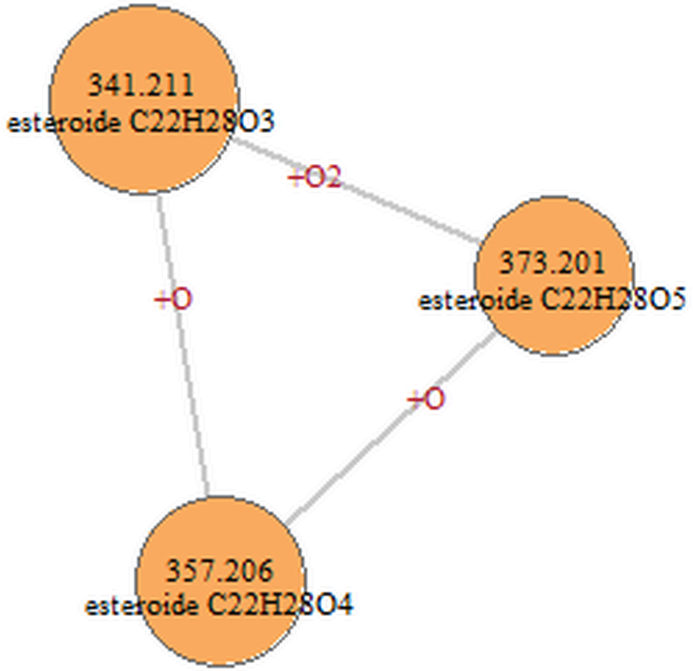


Negativo

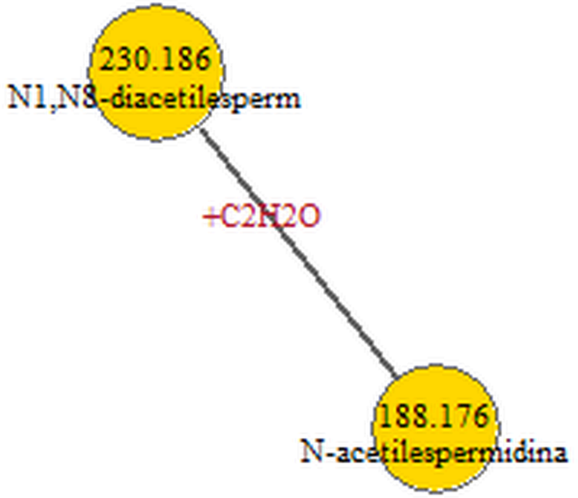
Red de transformaciones, positivo



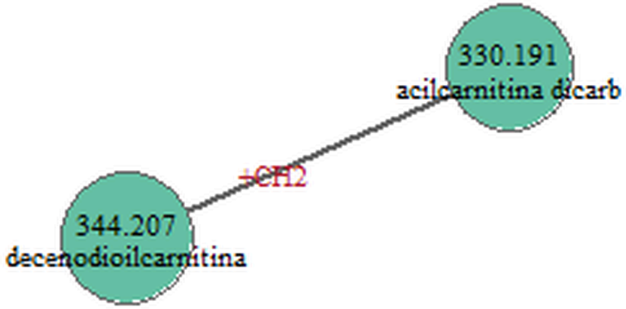
Fluoroquinolonas



Corticoides

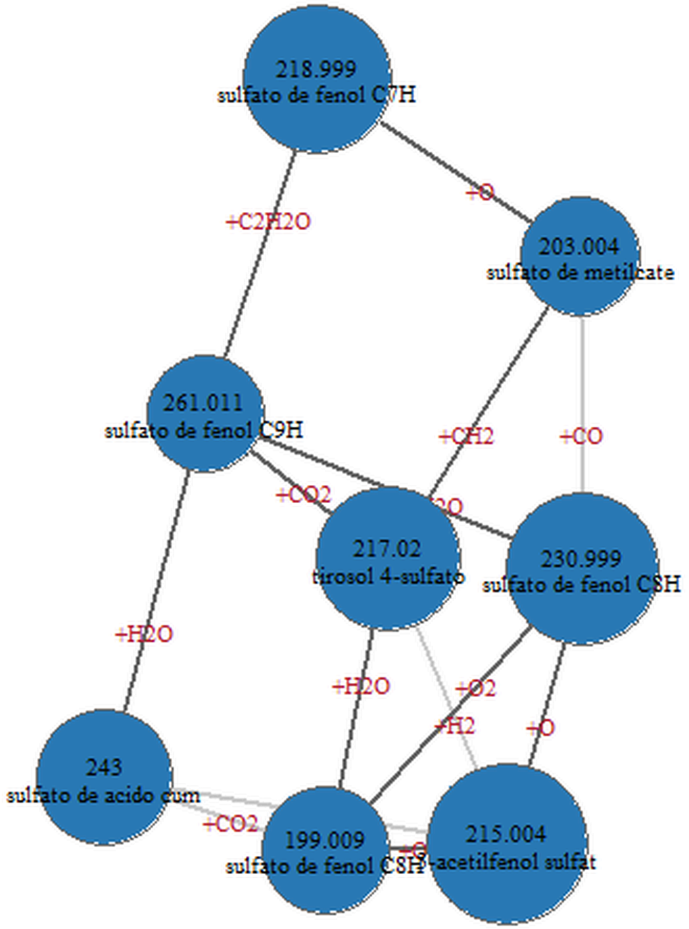


Poliaminas

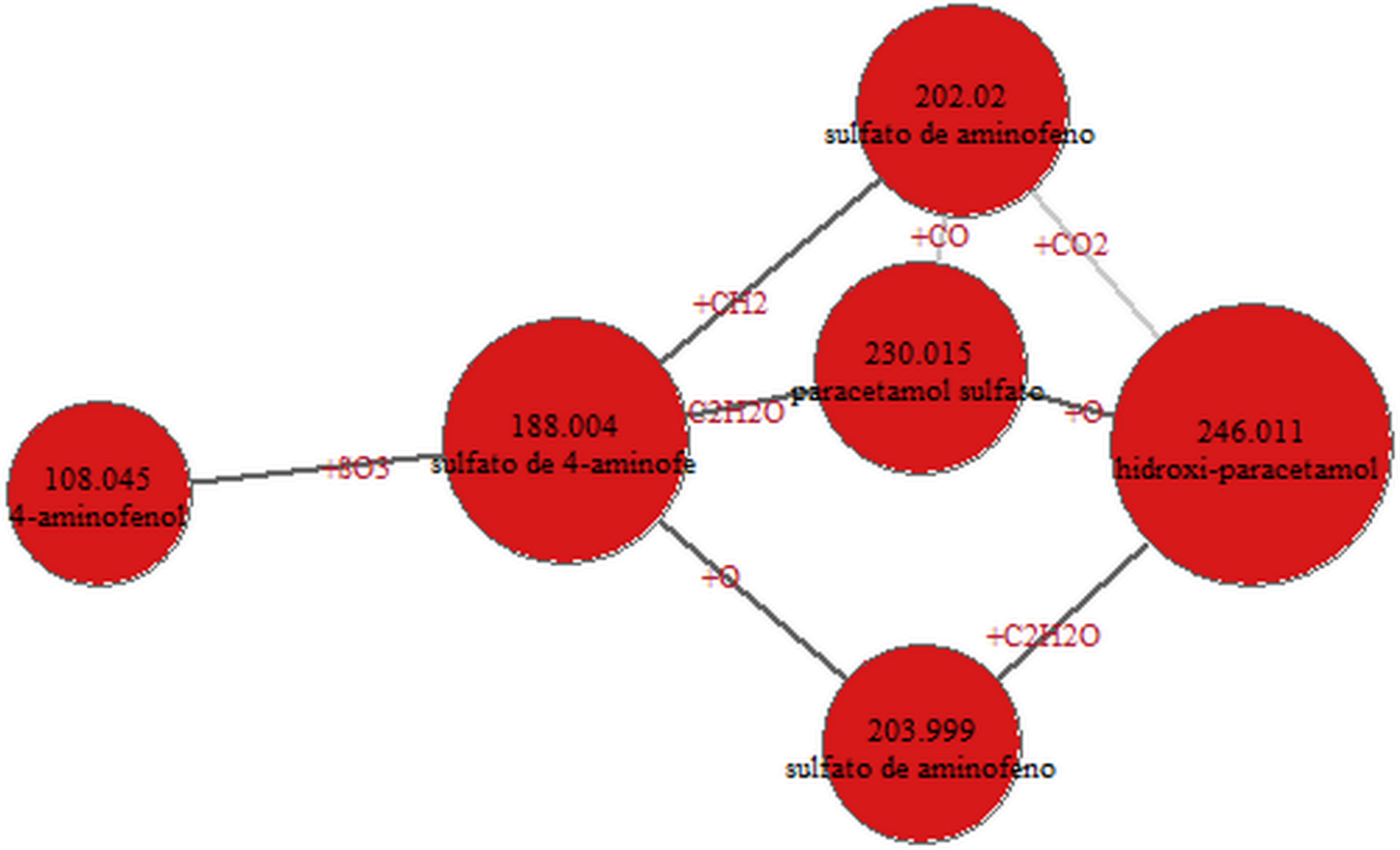


Acilcarnitinas

Red de transformaciones, negativo

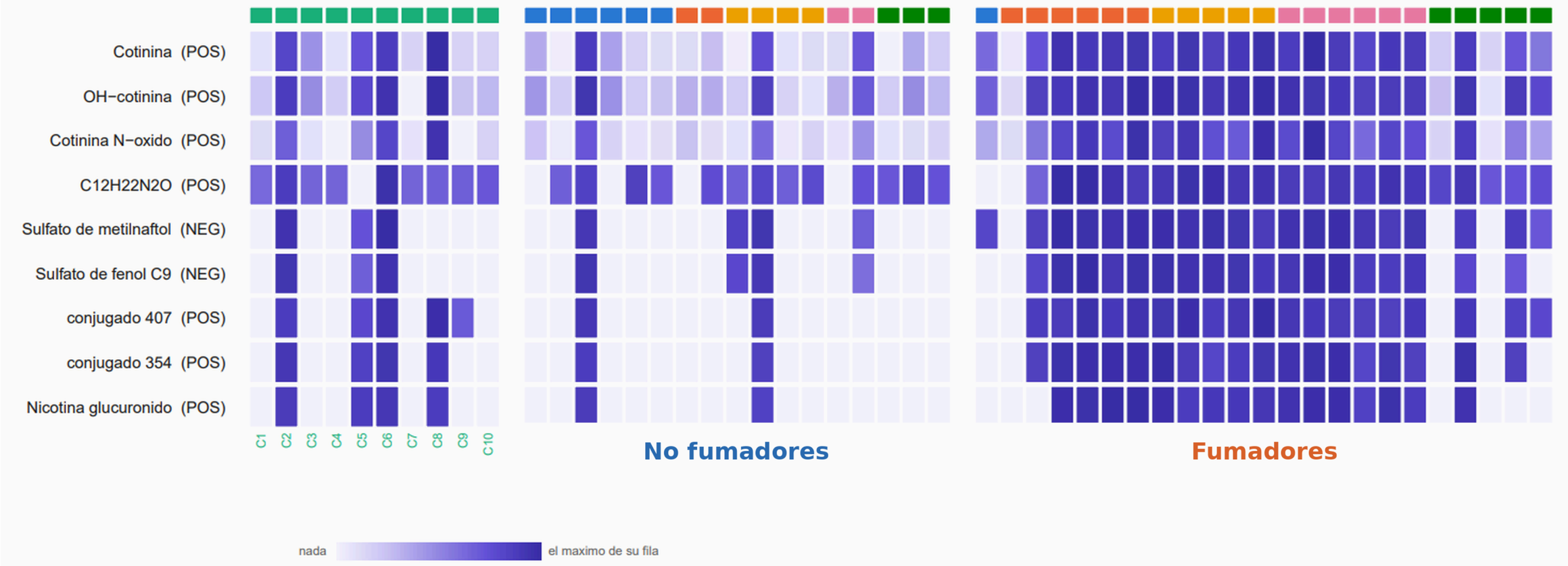


Sulfatos de fenol



Paracetamol y sus metabolitos

Marcadores de tabaco



Cotinina, OH-cotinina, nicotina glucurónido: casi todos los fumadores los tienen, casi ningún no fumador

Para el próximo estudio

- Pool QC hecho con las muestras del estudio
- Menos fondo, para perder menos features en el filtro de blancos
- Estándares internos que cubran todo el gradiente en negativo
- Controles con el mismo protocolo preanalítico
- Registro de medicación, tabaco y **dieta**

Análisis detallado: https://docs.google.com/document/d/15ql1KuiNT_3g52BBad5QgMKcRH8PJaTsCrmP3Ogv-AQ/edit?tab=t.0

Material de apoyo visual preparado con ayuda de Claude